

## Highlights

### **Emergent Composite Clustering and Gauge-Inspired Adaptive Connectivity in Agent-Based Neural Networks**

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- Static local phase covariance is proved for adaptive link transporters.
- Complete cycle-basis tests distinguish candidate from certified assemblies.
- Observable scale links enable a tested second level of composite assembly.
- Offset-redraw recovery is tested against frozen and algebraic controls.
- CNN weight sharing adds structure but not adversarial robustness.

# Emergent Composite Clustering and Gauge-Inspired Adaptive Connectivity in Agent-Based Neural Networks

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## Abstract

Neuronal assemblies are commonly modelled as strongly coupled, co-active units while heterogeneous connection phases remain fixed. We instead give each directed connection a learnable phase transporter and allow a static local rephasing of every oscillatory unit. The transporter follows a normalised covariant mismatch; coupling and frequency adapt separately. A candidate composite is certified only when reciprocal two-cycles and a fundamental cycle basis make its transported mean path independent. Linear cluster elimination is static-exact, whereas transients retain memory and two poles. In a synchronous model of 1000 units and 5 realisations, adaptive transport reduces the 95th-percentile cycle residual from 1.7064 to 0.0002 rad and recruits 4.0 times the delay-only candidate fraction. It does not improve the fine-scale planted-partition recovery: the adjusted Rand indices are 0.15 and 0.17. A separate held-out level supplies observable mean-frequency links and raises fixed- $k$  ARI from 0.163 to 0.763; gauge transport chiefly adds complete-cycle consistency, and the strict selector abstains in 2 of 5 runs. After all offsets are redrawn, active transporters retain coherence 0.99 and partition identity 0.99, against 0.53 and 0.10 for delay only. On CIFAR-10,

soft sharing compacts weights but reduces clean and adversarial accuracy. An exactly paired read-out interpolation verifies the phase-invariance control but gives an invariant-minus-noninvariant clean-accuracy difference of  $-0.2 \pm 1.3$  points, so invariance does not explain the phase-routing gain. The claims separate algebra, representation symmetry, confirmed predictions and falsified components.

*Keywords:* neuronal assemblies, gauge invariance, adaptive phase lags, composite neurons, agent-based model, Kuramoto–Stuart–Landau networks

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## 1. Introduction

Cell assemblies are a useful scale between individual neurons and whole brain areas: groups of units become strongly coupled and coordinated through plasticity [1, 2, 3, 4, 5]. Oscillatory assemblies also face heterogeneous transmission delays. At a fixed rhythm a delay appears as a link-specific phase offset, and distributed offsets can decide which units lock [7, 8, 6]. A second, logically distinct issue is representational: the zero of phase assigned to each unit is a coordinate convention. A model should distinguish invariance under a change of those coordinates from physical adaptation after a delay changes.

Gauge theory provides the appropriate graph construction for the first distinction: node-wise phase coordinates are related by link transporters, and closed-loop phase accumulation is coordinate independent [9, 10, 11]. We use that construction under static local rephasing and make the transporters adaptive. Partial compositeness provides a second algebraic pattern: an explicitly retained variable mixes with a collective mode and inherits a mediated response [12, 13]. We use a Schur complement for the static response

and retain the exact memory kernel and poles for transients.

*Why an additional model is useful.* A coupling-only adaptive network can weaken links that carry incompatible fixed offsets, but it has no link variable with which to compensate them. The proposed transporter supplies that variable. The model then makes a falsifiable dynamical claim: after a physical redraw of the offsets, transporter learning should preserve more of the coupling-defined assembly structure than matched controls. It also makes a mathematical claim: only a connected strong-coupling subgraph whose complete cycle constraints are small supports an approximately path-independent transported mean. These claims are tested separately from the exact static covariance and exact algebraic-compensator controls.

*Contributions.*

1. A neural graph model covariant under static unit-wise phase redefinitions, with adaptive directed transporters and closed-walk holonomy as the invariant connection diagnostic (Section 4).
2. One non-vacuous transporter gradient rule and two local invariant mismatch-driven rules for couplings and frequencies. Proposition 2 proves static covariance; Proposition 1 proves the existence of an exact algebraic compensator for any offset change. A deterministic negative control shows that arbitrary time-dependent rephasing is not a symmetry of the complete adaptive system.
3. A distinction between coupling-defined candidate clusters and certified approximately flat composites. Proposition 3 proves exact path independence on a flat subgraph, and the numerical diagnostic tests

reciprocal two-cycles and a complete fundamental cycle basis rather than triangles alone.

4. An exact fixed-point Schur complement and an exact transient memory kernel for an elementary variable coupled to a collective cluster mode. The full two-mode poles replace the incorrect instantaneous effective-time formula of the earlier draft (Section 5).
5. A synchronous implementation with unit and link agents, three comparison arms, diagnostic controls, physical offset redraw, unit removal and frequency-step tests, and predictions fixed in the dated revision log before the corrected production run (Sections 6 and 7).
6. A prospective held-out hierarchy test that promotes only certified fine composites, supplies an observable scale relation without planted labels, and separates the partition gain due to that information from the complete-cycle consistency due to gauge transport (Sections 6.9 and 7.4).
7. A bounded weight-space study on CIFAR-10: fixed and learned semi-composite regularisers and two routing layers are compared under one training protocol, with clean accuracy, adversarial accuracy, compactness and both inference and total trainable parameter counts reported (Section 8).

Section 2 places the work among adaptive-oscillator, gauge-equivariant and soft-sharing models. Section 3 fixes the terminology. Sections 4–5 give the theory, Sections 6–7 the agent-based experiment, and Section 8 the bounded convolutional counterpart. The Discussion and Conclusion state both the positive and negative results.

## 2. Related work

*Gauge equivariance in learned systems.* Gauge-equivariant convolutional networks treat the choice of local geometric frame as a coordinate freedom and constrain layers to commute with it [14]; lattice-gauge networks preserve the transformation law of physical link fields [15, 16, 17]. Here the local coordinate is the phase origin of an internal oscillator and the link transporter is itself adaptive. This is a different use of the same covariance construction. Complex-valued networks supply the phase-carrying state space [18], while learned internal frames have also been proposed at a conceptual level in neuroscience [19]. None of these results by itself implies robustness to an input-space adversarial attack or to a physical change of transmission offsets; those are separate experiments.

*Adaptive synchronization and phase lags.* The phase reduction of Eq. (1) is a Sakaguchi–Kuramoto network with link-specific lags [7]. Delays and distributed lags change synchronization regimes [8, 6]; adaptive couplings can create multiclusters [20, 21]; and natural frequencies may themselves adapt [22]. Lodi et al. derive existence and stability conditions for synchronized cluster patterns in broad adaptive multilayer networks, including delays and quotient reductions [23]. Their equitable-partition and identical-state conditions address exact cluster-synchronous solutions. Our transported synchronization instead allows non-zero link phase relations and diagnoses their consistency by loop residuals. Anwar et al. extend adaptive synchronization to higher-order interactions [24]; the present model remains pairwise and adds a learned link transporter. These comparisons locate the contribution in the transporter and cycle diagnostics, not in adaptive clustering itself.

*Frustration on graphs.* A non-zero sum of link phases around a loop is the frustration invariant of XY gauge-glass constructions [25, 26]. We borrow the loop observable and the distinction between pure-gauge and cycle degrees of freedom. Because our system is noisy, adaptive and out of equilibrium, we do not identify its detected clusters with thermodynamic phases of a gauge–Higgs model [27].

*Assemblies and intrinsic plasticity.* Hebbian and homeostatic mechanisms can form and maintain neural assemblies [4, 5, 28, 29]. Population oscillators and their Kuramoto reductions are established coarse-grained descriptions [30, 31, 32]. Our model asks a narrower question within that tradition: what changes when the phase lag on each link is an adaptive variable with an exact static-coordinate covariance?

*Soft weight sharing.* The convolutional experiment is closest to soft weight sharing, which pulls weights towards learned centres [33, 34]. Its semi-composite extension lets some output units use a learned mixture of centres. This is an architectural counterpart of mixing with collective modes; it is not a realization of the oscillator dynamics or evidence for its biological interpretation.

### 3. Definitions and notation

This section fixes the meaning of every framework-specific term. In particular, it separates a coupling-defined *candidate cluster* from a cluster that passes the full cycle-consistency test.

**Definition 1** (Network and agents). *The network is a graph with  $N$  unit agents and reciprocal directed link variables on every undirected support edge. A directed variable  $i \leftarrow j$  is owned by the receiving unit  $i$ . A unit reads its state and incoming links; a link reads the states of its two endpoints. This is the agent-based meaning used below.*

**Definition 2** (Unit state). *Unit  $i$  has complex state  $z_i(t) = r_i(t)e^{i\phi_i(t)}$ . The modulus  $r_i$  is the amplitude of oscillatory activity and  $\phi_i$  its phase. The Stuart–Landau dynamics used below is a population-level normal form near an oscillatory instability, not a single-neuron membrane model [30, 31].*

**Definition 3** (Intrinsic, or elementary, unit). *A unit is elementary when it is retained explicitly rather than replaced by a collective variable. In the linear reduction of Section 5, its bare coefficients are those of the full elementary–cluster system; the coefficients after eliminating the cluster are its effective response. In the simulation, a unit not assigned to a coupling-defined candidate cluster is labelled elementary.*

**Definition 4** (Static local phase transformation). *A static local phase transformation chooses one time-independent angle  $\alpha_i$  per unit and sends  $z_i \rightarrow e^{i\alpha_i}z_i$ . It changes the phase coordinates, not the physical state. A quantity unchanged by every such transformation is invariant; one multiplied by  $e^{i\alpha_i}$  is covariant at  $i$ . The full adaptive system below is claimed to have this static symmetry only.*

**Definition 5** (Transporter, offset and total connection). *A directed link  $i \leftarrow j$  has a fixed transmission offset  $\delta_{ij}$  and a learnable transporter phase  $\theta_{ij}$ . Their sum defines  $U_{ij} = e^{i(\theta_{ij} + \delta_{ij})}$ . Under Definition 4,  $\theta_{ij} \rightarrow \theta_{ij} + \alpha_i - \alpha_j$ ,*

while  $\delta_{ij}$  is unchanged. The offset is a property of the transmission medium; the transporter is an adaptive model variable.

**Definition 6** (Covariant difference and invariant link correlation). *The covariant difference on  $i \leftarrow j$  is  $z_i - U_{ij}z_j$ . With  $\hat{z}_i = z_i/|z_i|$ , the invariant link correlation is  $m_{ij} = \bar{\hat{z}}_i U_{ij} \hat{z}_j$ . The transporter rule uses  $\text{Im } m_{ij}$ , whereas the coupling and frequency rules use invariant functions of  $m_{ij}$ .*

**Definition 7** (Adaptation sensitivity and natural frequency). *The positive coefficient  $q_i$  multiplies the transporter update on links received by unit  $i$  and is therefore an adaptation sensitivity. It is not a representation-theoretic charge, and no additive cluster charge is assumed. The natural frequency  $\omega_i$  is the unit's isolated angular frequency and is itself adapted by an invariant local rule.*

**Definition 8** (Cycle residual, flatness and reversibility defect). *For a directed closed walk  $\gamma$ , the loop holonomy is the wrapped sum  $\Phi_\gamma = \sum_{(i \leftarrow j) \in \gamma} (\theta_{ij} + \delta_{ij}) \bmod 2\pi$ . A connected subgraph is flat when every directed loop has zero holonomy. Numerically, flatness is tested by every reciprocal two-cycle and by a fundamental cycle basis; a cluster is approximately flat at tolerance  $\varepsilon_\Phi$  when the absolute residual of every such constraint is at most  $\varepsilon_\Phi$ . The reciprocal two-cycle residual is the reversibility defect  $(\theta_{ij} + \delta_{ij}) + (\theta_{ji} + \delta_{ji}) \bmod 2\pi$ .*

**Definition 9** (Candidate cluster, certified composite and transported operator). *A candidate cluster is a connected component, of at least  $n_{\min}$  units, of the symmetrised learned-coupling graph above a stated threshold. It is a certified composite cluster only if it also passes Definition 8. For a certified cluster*

$C$ , the transported operator read at member  $i$  is  $V_C^{(i)} = |C|^{-1} \sum_{k \in C} U_{ik}^{(\gamma_k)} z_k$ , where  $U_{ik}^{(\gamma_k)}$  is the product of link connections along a path from  $k$  to  $i$ . Proposition 3 gives the condition under which this operator is path independent. The simulation reports candidates, their connectivity coverage and the fraction that are certified; it does not silently equate the thresholded coupling component with an exact composite.

**Definition 10** (Effective response). *An effective response is the equation for retained variables after another set of variables has been eliminated. At a fixed point it is an ordinary Schur complement. For time-dependent dynamics the exact response can depend on Laplace frequency, equivalently on the history of the retained variables. A constant “effective time” is therefore meaningful only as a specified approximation or as an exact pole of the full linear system.*

## 4. Gauge-inspired structure on a neural graph

### 4.1. Static local phase covariance and algebraic compensation

The phase assigned to an oscillatory unit depends on a reference. If each unit is given its own static reference shift, a comparison between neighbouring phases must transport one phase into the other’s frame. Definition 5 supplies that transporter. This is an exact covariance of the equations under time-independent local rephasings; it is not a claim that every physical change of conduction delay is a gauge transformation.

**Proposition 1** (Algebraic compensation of an offset change). *The state equation and the three adaptation rules depend on  $\theta_{ij}$  and  $\delta_{ij}$  only through their sum. For any physical offset change  $\delta_{ij} \rightarrow \delta_{ij} + \Delta_{ij}$ , the simultaneous*

Table 1: Symbols and their transformation under the static local rephasing of Definition 4.

| Symbol                               | Meaning                                                             | Transformation                                  |
|--------------------------------------|---------------------------------------------------------------------|-------------------------------------------------|
| $z_i = r_i e^{i\phi_i}$              | complex state of unit $i$                                           | $e^{i\alpha_i} z_i$                             |
| $\omega_i$                           | adapted natural frequency                                           | invariant                                       |
| $q_i$                                | transporter adaptation sensitivity                                  | invariant                                       |
| $\mu, D$                             | Stuart–Landau growth and noise parameters                           | invariant                                       |
| $J_{ij} \in [0, J_0]$                | directed coupling strength                                          | invariant                                       |
| $\delta_{ij}$                        | fixed transmission offset                                           | invariant                                       |
| $\theta_{ij}$                        | learnable transporter phase                                         | $\theta_{ij} + \alpha_i - \alpha_j$             |
| $U_{ij}$                             | total link connection                                               | $e^{i\alpha_i} U_{ij} e^{-i\alpha_j}$           |
| $m_{ij}$                             | transported phase correlation                                       | invariant                                       |
| $\Phi_\gamma$                        | closed-walk holonomy                                                | invariant                                       |
| $\varepsilon_\Phi$                   | numerical flatness tolerance                                        | invariant                                       |
| $C, V_C^{(i)}$                       | candidate/certified cluster and transported operator                | $V_C^{(i)} \rightarrow e^{i\alpha_i} V_C^{(i)}$ |
| $\Lambda, \epsilon_i, g_C$           | intra-cluster coupling, mixing coefficient, static cluster response | invariant                                       |
| $R_C^{\text{raw}}, R_C^{\text{cov}}$ | raw and coverage-aware transported coherence                        | only $R_C^{\text{cov}}$<br>invariant            |

*algebraic change  $\theta_{ij} \rightarrow \theta_{ij} - \Delta_{ij}$  leaves every  $U_{ij}$ , and hence the complete vector field and every reported invariant, unchanged.*

The proposition is an existence statement: a compensating transporter configuration exists. An offset redraw in the numerical test changes  $\delta$  while leaving  $\theta$  untouched, so it is a genuine perturbation. Whether the local dynamics finds a compensating state is measured rather than inferred. We separate three controls: exact static covariance under a change of phase coordinates; the algebraic compensator of Proposition 1; and recovery after a physical offset redraw, including a gauge-network control in which the learned transporters are frozen. The proof is in Appendix [Appendix A](#).

#### 4.2. Dynamics

Each unit is a Stuart–Landau oscillator coupled through covariant differences,

$$\dot{z}_i = \left(\mu + i\omega_i - |z_i|^2\right) z_i + \sum_j J_{ij} (U_{ij} z_j - z_i) + \sqrt{2D} \xi_i(t), \quad (1)$$

where  $\xi_i$  is complex white noise of unit intensity. Under a static local rephasing, every deterministic term at unit  $i$  acquires the same factor  $e^{i\alpha_i}$ . We do not claim covariance under arbitrary time-dependent rephasings: the adaptation laws for  $\theta$ ,  $J$  and  $\omega$  do not supply the additional connection terms required for that larger symmetry. The supplementary negative control makes this boundary explicit.

When amplitudes remain close to  $\sqrt{\mu}$ , Eq. (1) reduces to

$$\dot{\phi}_i = \omega_i + \sum_j J_{ij} \sin(\phi_j - \phi_i + \theta_{ij} + \delta_{ij}), \quad (2)$$

a Sakaguchi–Kuramoto network with a link-specific adaptive lag [7].

#### 4.3. One gradient rule and two invariant mismatch-driven rules

The normalised link mismatch is  $|\hat{z}_i - U_{ij}\hat{z}_j|^2 = 2 - 2 \operatorname{Re} m_{ij}$ . Its sum

$$\hat{E} = \sum_{(i \leftarrow j) \in \mathcal{E}} |\hat{z}_i - U_{ij}\hat{z}_j|^2 \quad (3)$$

is invariant under static rephasing. Direct differentiation gives  $\partial \hat{E} / \partial \theta_{ij} = 2 \operatorname{Im} m_{ij}$ , so transporter adaptation is the gradient flow

$$\dot{\theta}_{ij} = -2\eta_\theta q_i \operatorname{Im} m_{ij}. \quad (4)$$

Here  $q_i$  is the receiving unit's adaptation sensitivity. The production model uses the normalised mismatch so that transporter correction does not stop merely because a coupling has already weakened. The code also supplies the  $J$ -weighted alternative as a sensitivity check.

The remaining two updates are local invariant rules driven by the same mismatch, but they are not derived as gradients of Eq. (3):

$$\dot{J}_{ij} = \eta_J (J_0 \operatorname{Re} m_{ij} - J_{ij}), \quad J_{ij} \in [0, J_0], \quad (5)$$

$$\dot{\omega}_i = \eta_\omega \sum_j J_{ij} \operatorname{Im} m_{ij}. \quad (6)$$

Equation (5) is a transported Hebbian correlation with decay and clipping. Equation (6) is the adaptive-frequency mechanism of Taylor, Ott and Restrepo written with the invariant correlation [22]. Its role is operational: it reduces persistent local phase mismatch on the coupling-defined neighbourhood. It is not needed to establish the static symmetry.

**Proposition 2** (Static covariance). *Under every time-independent local phase transformation of Definition 4, Eq. (1) and Eqs. (4)–(6) are covariant, while  $m_{ij}$ ,  $\hat{E}$ , every closed-walk holonomy, every cycle-consistency residual and every coverage-aware transported coherence are invariant.*

**Remark 1** (Earlier vacuous objective). *The previous manuscript proposed descent on  $|D_t V_i - (\partial_t - i q_i A_{ij}) V_j|^2$  with  $D_t V_i = \partial_t V_i - i q_i \sum_k A_{ik} V_k$ . The explicit  $A_{ij} V_j$  contribution cancels the same term in the sum, so the residual is independent of  $A_{ij}$  and the stated gradient vanishes identically. The supplementary symbolic check proves the cancellation. Equation (4) is the non-vacuous replacement.*

#### 4.4. Loop field strength and frustration

For any directed closed walk  $\gamma$ ,

$$\Phi_\gamma = \sum_{(i \leftarrow j) \in \gamma} (\theta_{ij} + \delta_{ij}) \bmod 2\pi \quad (7)$$

is invariant. Testing triangles alone is insufficient on a general graph: a triangle-free square may carry non-zero holonomy. The numerical test therefore reconstructs a spanning-tree potential in every candidate cluster and evaluates all reciprocal two-cycles and every non-tree edge, which forms a complete fundamental cycle basis. The deterministic negative control is a triangle-free square with a non-zero loop residual.

Fixed random offsets create the same loop frustration that appears in XY gauge-glass models [25, 26]. This is a structural correspondence of link phases and loop invariants, not a claim that the neural dynamics realises the equilibrium thermodynamics or phases of those models. Likewise, a learned approximately flat strongly coupled subgraph resembles the ordered sector of a lattice gauge-matter system [27]; we use the correspondence to define observables and controls, not to identify a neural phase transition.

Table 2: Gauge-theory constructions used here and their restricted neural meanings. The last two rows are structural correspondences, not thermodynamic identifications.

| Construction                | Meaning in this model                              | Location         |
|-----------------------------|----------------------------------------------------|------------------|
| Static local transformation | independent change of phase coordinates            | Definition 4     |
| Matter-like variable        | complex unit state $z_i$                           | Eq. (1)          |
| Spatial connection          | learnable transporter plus fixed offset            | Definition 5     |
| Covariant difference        | comparison after link transport                    | Definition 6     |
| Loop field strength         | closed-walk holonomy and cycle residual            | Eq. (7)          |
| Flat connection             | all independent cycle constraints vanish           | Definition 8     |
| Gauge-glass correspondence  | fixed disordered link offsets and loop frustration | Section 4.4      |
| Gauge-matter correspondence | adaptive states and link phases on a graph         | Sections 4.2–4.3 |
| Composite operator          | transported cluster mean when path independent     | Proposition 3    |
| Partial compositeness       | elementary variable coupled to a collective mode   | Section 5.2      |

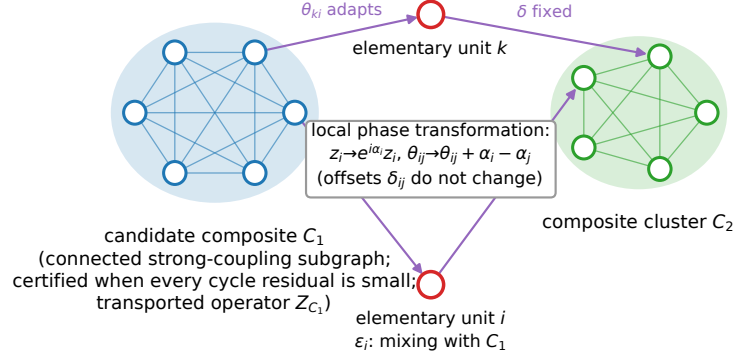


Figure 1: Candidate composite clusters, elementary units and directed links. A candidate is a connected component of the thresholded learned-coupling graph. It is certified as approximately flat only after all reciprocal and fundamental-cycle residuals pass the stated tolerance. On an exactly flat cluster the transported operator is path independent. The coefficient  $\epsilon_i$  in Section 5.2 couples a retained elementary variable to the collective cluster mode.

## 5. Composite neurons and partial compositeness

### 5.1. From a coupling-defined candidate to a composite operator

Figure 1 shows the objects used in the construction. Strong coupling alone defines a candidate; cycle consistency supplies the additional condition needed for a single transported collective variable.

**Proposition 3** (Path independence and flatness). *Let  $C$  be connected and let  $U_{ik}^{(\gamma)}$  be the product of directed connections along a path  $\gamma$  from  $k$  to  $i$ . The transported mean  $V_C^{(i)} = |C|^{-1} \sum_{k \in C} U_{ik}^{(\gamma_k)} z_k$  is independent of every path choice if and only if all directed loop holonomies in  $C$  vanish. Under a static rephasing it is covariant at  $i$ , whereas the untransported mean is not.*

The proof is in [Appendix A](#). In finite-precision simulations we test a complete cycle basis rather than triangles alone and use “certified composite” only when every tested residual is below  $\varepsilon_\Phi$ . Otherwise the object remains a candidate cluster and its transported coherence is reported together with the fraction of members covered by the connected component used for transport.

### 5.2. Static Schur complement and exact transient response

The partial-compositeness calculation is a linear response result. Work in a frame in which an exactly flat cluster has unit transport on its internal links and consider

$$\tau \dot{V}_i = -V_i + \sum_j J_{ij} V_j + I_i. \quad (8)$$

A cluster  $C$  contains  $n$  units with uniform off-diagonal coupling  $\Lambda$ . An elementary variable  $V_i$  couples symmetrically to every cluster member with coefficient  $\epsilon_i$ . For the cluster mean  $V_C = n^{-1} \sum_{k \in C} V_k$  and mean input  $I_C$ ,

$$\tau \dot{V}_C = -a V_C + \sum_{j \notin C} \epsilon_j V_j + I_C, \quad a = 1 - (n-1)\Lambda. \quad (9)$$

The uniform cluster mode is stable when  $a > 0$ , or  $\Lambda < \Lambda_c = 1/(n-1)$ .

At a fixed point, eliminating  $V_C$  gives the exact Schur complement

$$J_{EE}^{\text{eff}}(0) = J_{EE} + g_C \boldsymbol{\epsilon} \boldsymbol{\epsilon}^\top, \quad g_C = \frac{n}{a}, \quad (10)$$

with static self-feedback and mediated coupling

$$\Sigma_i(0) = g_C \epsilon_i^2, \quad (11)$$

$$J_{ij}^{\text{eff}}(0) = J_{ij} + g_C \epsilon_i \epsilon_j, \quad (12)$$

$$I_i^{\text{eff}}(0) = I_i + g_C \epsilon_i I_C. \quad (13)$$

If a bare threshold enters  $I_i$  with a minus sign, the last equation is the corresponding static threshold shift. The supplementary check verifies Eq. (10) and the full fixed point to machine precision and uses the plain cluster mean without the recurrent factor  $g_C$  as a negative control.

For transients, substituting the fixed-point value of  $V_C$  into a differential equation is not exact. Solving Eq. (9) gives

$$V_C(t) = e^{-at/\tau} V_C(0) + \frac{1}{\tau} \int_0^t e^{-a(t-s)/\tau} \left[ \sum_j \epsilon_j V_j(s) + I_C(s) \right] ds, \quad (14)$$

so eliminating the cluster produces a memory kernel. In Laplace frequency,

$$J_{EE}^{\text{eff}}(s) = J_{EE} + \frac{n \boldsymbol{\epsilon} \boldsymbol{\epsilon}^T}{a + \tau s}. \quad (15)$$

Equation (10) is the exact zero-frequency limit of Eq. (15); it is not an exact instantaneous evolution law.

For one elementary variable and the uniform cluster mode, the exact dynamics is

$$\tau \frac{d}{dt} \begin{pmatrix} V_i \\ V_C \end{pmatrix} = \begin{pmatrix} -1 & n\epsilon_i \\ \epsilon_i & -a \end{pmatrix} \begin{pmatrix} V_i \\ V_C \end{pmatrix} + \begin{pmatrix} I_i \\ I_C \end{pmatrix}. \quad (16)$$

Its poles and slow relaxation time are

$$\lambda_{\pm} = \frac{-(1+a) \pm \sqrt{(1-a)^2 + 4n\epsilon_i^2}}{2\tau}, \quad (17)$$

$$\tau_{\text{slow}} = -\lambda_+^{-1} \quad (\lambda_+ < 0). \quad (18)$$

The pair becomes marginal at

$$\Lambda^* = \frac{1 - n\epsilon_i^2}{n - 1} < \Lambda_c, \quad (19)$$

where the determinant vanishes. Beyond this point the linear pair is unstable; the calculation alone does not establish absorption into a new nonlinear branch.

The distinction is numerically substantial. For  $n = 6$  and  $\epsilon = 0.2$ , at  $\Lambda = 0.10$  the commonly used static-local estimate  $\tau/(1 - g_C\epsilon^2)$  is  $1.92\tau$ , a first-order low-frequency expansion gives  $3.77\tau$ , and the exact slow pole is  $5.00\tau$ . At  $\Lambda = 0.14$  the three values are  $5.00\tau$ ,  $18.33\tau$  and  $20.87\tau$ . We therefore use the Schur complement only for static gains and couplings and use Eq. (17) for transient time scales.

### 5.3. Correspondence and limits

The particle-physics comparison is an algebraic correspondence: an explicit variable mixes with a collective mode and inherits a mediated static interaction and a modified pole structure [12, 13]. It does not imply Lorentz invariance, fermion statistics, renormalisation, an additive neural charge or a composite-Higgs phase.

## 6. Agent-based implementation

Every unit and every directed link carries its own state and reads only local information. Global objects such as candidate clusters are diagnostics computed after the local dynamics; they are not supplied to the agents. This is the sense in which the model is agent based [35].

### 6.1. Agents and local variables

Unit  $i$  stores  $(z_i, \omega_i, q_i)$  and its incoming links. A directed link  $i \leftarrow j$  stores  $(J_{ij}, \theta_{ij}, \delta_{ij})$ . Unit dynamics is Eq. (1); links use Eqs. (4) and (5); units

Table 3: Partial compositeness and the restricted linear-response counterpart used here.

| Particle-physics<br>structure   | Neural linear-response construction                           | Location |
|---------------------------------|---------------------------------------------------------------|----------|
| elementary degree of<br>freedom | retained variable $V_i$                                       | Eq. (8)  |
| composite operator              | transported cluster mean when path<br>independent             | Prop. 3  |
| mixing parameter                | elementary–cluster coefficient $\epsilon_i$                   | Eq. (16) |
| strong-sector response          | static cluster susceptibility $g_C = n/a$                     | Eq. (10) |
| mediated interaction            | $J_{ij}^{\text{eff}}(0) = J_{ij} + g_C \epsilon_i \epsilon_j$ | Eq. (12) |
| frequency-dependent<br>dressing | memory kernel $n \epsilon_i \epsilon_j / (a + \tau s)$        | Eq. (15) |
| physical time scale             | pole $\tau_{\text{slow}} = -1/\lambda_+$                      | Eq. (18) |
| stability boundary              | zero determinant at $\Lambda^*$                               | Eq. (19) |

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**Algorithm 1** One synchronous step; every right-hand side uses step- $s$  values.

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1: for all directed links  $i \leftarrow j$  do
2:    $U_{ij}^s \leftarrow e^{i(\theta_{ij}^s + \delta_{ij})}$ ;  $m_{ij}^s \leftarrow \bar{z}_i^s U_{ij}^s \hat{z}_j^s$ 
3:   form  $d\theta_{ij}^s = -2\eta_\theta q_i \text{Im } m_{ij}^s$  for the gauge arm, and  $dJ_{ij}^s =$ 
      $\eta_J(J_0 \text{Re } m_{ij}^s - J_{ij}^s)$ 
4: end for
5: for all units  $i$  do
6:   form  $dz_i^s = (\mu + i\omega_i^s - |z_i^s|^2)z_i^s + \sum_j J_{ij}^s(U_{ij}^s z_j^s - z_i^s)$ 
7:   form  $d\omega_i^s = \eta_\omega \sum_j J_{ij}^s \text{Im } m_{ij}^s$ 
8: end for
9: assign  $\theta^{s+1} = \theta^s + \Delta t d\theta^s$ ,  $J^{s+1} = \text{clip}(J^s + \Delta t dJ^s, 0, J_0)$ 
10: assign  $\omega^{s+1} = \omega^s + \Delta t d\omega^s$ ,  $z^{s+1} = z^s + \Delta t dz^s + \sqrt{2D\Delta t} \xi^s$ 

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update  $\omega_i$  by Eq. (6). Candidate clusters are detected only after learning.

### 6.2. Synchronous implementation

The simulation uses synchronous Euler–Maruyama. At step  $s$ , all  $U_{ij}$ ,  $m_{ij}$  and derivatives are formed from  $(z^s, \theta^s, J^s, \omega^s)$ . Only after every derivative has been formed are the four arrays assigned their step- $s+1$  values. Thus no agent reads another agent’s partially updated state. We do not claim equivalence to asynchronous scheduling; the numerical results refer to Algorithm 1 only.

### 6.3. Graph, planted groups and initial state

Positions are sampled uniformly in the unit square. Two units share an undirected support edge when their Euclidean distance is below  $r = \sqrt{k/(\pi N)}$ ; every support edge carries both directed link variables. Thus

$k$  is a target interior degree, while realised degrees vary and boundary units have fewer neighbours. The same support, positions and stochastic seed are used in all arms.

Each unit is assigned, for scoring only, to one of  $G$  planted frequency groups. Initial frequencies are  $\Omega_g + \sigma_\omega \zeta_i$ , where  $\Omega_g = \Delta\Omega[g - (G - 1)/2]$  and  $\zeta_i$  is standard normal. Directed offsets are independent uniform draws from  $[-\delta_{\max}, \delta_{\max}]$ . Initially  $J_{ij} = J_0$ ,  $\theta_{ij} = 0$  in the gauge and delay-only arms,  $q_i = 1$ , and  $z_i = 0.1(x_i + iy_i)$  with independent standard-normal  $x_i, y_i$ . These choices match the released code and result metadata.

#### 6.4. Comparison arms

- **Gauge arm:** transporters, couplings and frequencies all adapt.
- **Delay-only arm:**  $\theta_{ij} = 0$  remains fixed;  $J$  and  $\omega$  adapt.
- **Quenched-transporter arm:** each  $\theta_{ij}$  is drawn uniformly once and then held fixed;  $J$  and  $\omega$  adapt.

The latter two controls distinguish learned transport from fixed heterogeneous phase lags. Their structural relation to gauge-glass models is the loop-frustration correspondence of Section 4.4, not a thermodynamic identity.

#### 6.5. Measurements

A candidate cluster is a connected component of the undirected graph for which the symmetrised coupling  $(J_{ij} + J_{ji})/2$  exceeds  $J_0/2$ , with at least  $n_{\min}$  members. We report the assigned fraction, adjusted Rand index (ARI) against the planted groups [36], size-weighted purity and clusters per planted group.

For a candidate  $C$ , raw coherence is  $R_C^{\text{raw}} = |C|^{-1} |\sum_{k \in C} \hat{z}_k|$ . Transported coherence is computed only on a connected strong-coupling component: states are transported to a spanning-tree root. If a tracked candidate fragments after a perturbation, we report the largest-component value  $R_{C,\text{lcc}}^{\text{cov}}$ , its membership coverage  $f_{C,\text{lcc}}$ , and the conservative score

$$R_{C,\text{covg}}^{\text{cov}} = f_{C,\text{lcc}} R_{C,\text{lcc}}^{\text{cov}}. \quad (20)$$

A disconnected set therefore cannot score one by silently omitting members.

Flatness is evaluated on every reciprocal two-cycle and every fundamental cycle generated by non-tree edges. We report the median, 95th percentile and maximum absolute residual, the fraction of constraints below  $\varepsilon_\Phi$ , and the fraction of candidate-assigned units belonging to clusters whose *maximum* residual passes the tolerance. This complete test replaces the triangle-only diagnostic of the previous draft. Frequency spread, coupling strengths and late transporter drift are also reported.

### 6.6. Controls

The released checks contain known-true and known-false cases: a connected flat graph must pass; a triangle-free square with non-zero loop holonomy must fail; and a disconnected five-node set with a coherent three-node component must receive coverage  $3/5$ , not full coherence. For every simulated gauge realisation, random static rephasing must leave the complete invariant diagnostic vector unchanged below  $10^{-9}$ , and the transporter gradient must be non-zero away from a stationary point. After an offset redraw, the algebraic assignment  $\theta \rightarrow \theta - (\delta_{\text{new}} - \delta_{\text{old}})$  must reproduce the old  $U$  to

machine precision. These controls test the mathematics separately from the network’s ability to relearn.

### 6.7. *Perturbation tests*

After an organisation interval  $T_1$ , three copies of each organised state are followed for  $T_2$ . In the *offset-redraw* test every directed  $\delta_{ij}$  is redrawn while  $\theta$ ,  $J$ ,  $\omega$  and  $z$  are initially left unchanged. We track the old candidates with Eq. (20), redetect candidates at the end and compute the before–after ARI. For the gauge arm we also rerun the perturbation with transporter learning frozen, while  $z$ ,  $J$  and  $\omega$  retain their ordinary updates; this isolates the contribution of relearning  $\theta$ . In the *removal* test, a random fraction of units is deleted and coverage-aware coherence of the survivors is measured. In the *frequency-step* test, half the units of planted group 0 receive an additive frequency step and their raw re-locking and residual frequency difference are measured.

### 6.8. *Predictions fixed before the corrected production run*

The corrected diagnostics make four claims falsifiable. (P1) Gauge-arm candidates have smaller complete cycle-basis residuals and a larger certified-flat fraction than both controls. (P2) Adaptive transport recruits a larger fraction of units into strong-coupling candidates and yields a higher adjusted Rand index with the planted frequency groups than both controls. (P3) Following a physical offset redraw, the gauge arm preserves more coverage-aware coherence and partition identity than the controls; freezing its learned transporters weakens that recovery. (P4) The static covariance and algebraic-compensator controls pass to machine precision. P1–P4 were written into the

dated revision log before the corrected production run; they were not publicly preregistered, and all outcomes are reported.

### 6.9. Prospective held-out extension to a second information scale

The fine-scale P2 test asks the local dynamics to recover a large planted group from a sparse graph without giving it an observable that identifies separated fragments of that group. Its negative outcome motivates a distinct question: when an observable relation at the next scale is available, can the gauge construction promote certified fragments and glue them into a valid larger composite? This is a separately registered extension on held-out seeds, not a redefinition of P2 after seeing its result.

Only fine candidates that pass every complete-cycle constraint at tolerance  $\varepsilon_\Phi$  are promoted. For a certified candidate  $C$ , choose a root  $r_C$  and let  $\psi_i^C$  be the tree transport from  $i$  to that root. Its promoted state and mean adapted frequency are

$$Z_C(t) = \frac{1}{|C|} \sum_{i \in C} e^{-i\psi_i^C} \hat{z}_i(t), \quad \bar{\omega}_C = \frac{1}{|C|} \sum_{i \in C} \omega_i. \quad (21)$$

Under a static fine-node rephasing,  $Z_C$  transforms by the phase at  $r_C$  and  $\bar{\omega}_C$  is invariant. A fine boundary link  $i \leftarrow j$ , with  $i \in C$  and  $j \in D$ , is dressed into the root frames as

$$\phi_{CD}^{(ij)} = -\psi_i^C + (\theta_{ij} + \delta_{ij}) + \psi_j^D, \quad (22)$$

which transforms as a transporter from the root of  $D$  to the root of  $C$ . The deterministic supplementary check verifies both transformation laws and includes a negative control.

The information that proposes new cross-fragment comparisons is kept separate from the transporter. During the final observation window we record the promoted states and form

$$M_{CD} = \frac{1}{L} \sum_{\ell=1}^L \frac{Z_C(t_\ell)}{|Z_C(t_\ell)|} \frac{\overline{Z_D(t_\ell)}}{|Z_D(t_\ell)|}. \quad (23)$$

The invariant mean-frequency difference  $|\bar{\omega}_C - \bar{\omega}_D|$  supplies the scale-level proximity, while  $\arg M_{CD}$  supplies a covariant initial phase for a proposed coarse link. The proposed support is the union of physical boundary pairs and the symmetric  $k$ -nearest graph in mean frequency. No planted label enters this construction. The coarse states and links then follow the same local dynamics as Eq. (1) and Eqs. (4)–(6), with slower declared rates.

The parameter is  $k \in \{2, 4, 8\}$ . A development seed fixed the label-free rule before the held-out run: choose the largest tested  $k$  for which the gauge-arm 95th-percentile complete-cycle residual is at most  $\varepsilon_\Phi = 0.01$  rad and at least 95% of promoted units lie in fully certified coarse modules; if no  $k$  passes, retain physical boundary support only. The matched no-transporter arm receives exactly the same selected support. Equal-count random links test whether any sparse augmentation suffices, and links chosen with the planted labels form a positive oracle control. Planted labels are opened only after selection to compute ARI and purity on seeds 10–14. Thus a gain shared by the two information-matched arms belongs to the supplied scale information, whereas a difference in complete-cycle consistency tests the additional role of gauge transport. Two levels demonstrate hierarchical composition; they do not by themselves establish a universal scaling law.

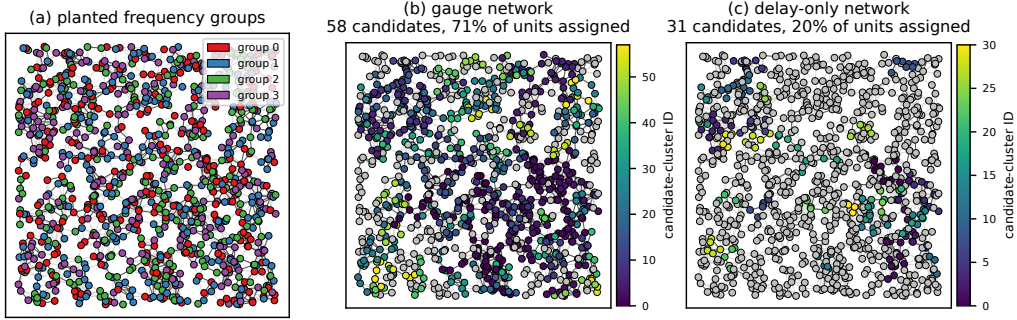


Figure 2: One realisation on the actual random-geometric positions used by the simulation. Light and dark links are below and above the candidate threshold  $J_0/2$ . (a) Planted frequency groups, which the learning rules do not observe. (b) Gauge-arm candidates. (c) Delay-only candidates. Grey nodes are unassigned; the continuous colour bar labels candidate identity and avoids reusing a categorical colour for different candidates.

## 7. Results of the agent-based model

All results below come from the corrected production runs of Algorithm 1 with  $N = 1000$  units and 5 independent realisations. Each realisation redraws the geometric graph, transmission offsets, initial states and dynamical noise. The parameters are listed in Appendix B, and every entry in the tables is generated directly from the archived JSON files by the supplementary script described in Section 11. The deterministic controls of Section 6.6 returned PASS in every realisation. Table 4 reports the state at the end of learning, Table 5 reports the perturbations, and Figures 2–4 show one realisation.

### 7.1. Candidate assemblies and coverage-aware coherence

The gauge arm assigns 0.76 of the units to 59.6 candidate clusters on average, compared with 0.19 in 28.8 candidates for the delay-only arm and 0.21

Table 4: State at  $t = T_1$  (mean  $\pm$  standard deviation over 5 realisations). A candidate cluster is a connected component of the thresholded, symmetrised coupling graph with at least 5 units. Coverage is the fraction of a candidate lying in the largest component of the reciprocal directed support used by the transported coherence. Complete flatness is tested on reciprocal two-cycles and a fundamental cycle basis, with tolerance 0.01 rad.

| Quantity                                                               | Gauge network       | Delay-only network  | Quenched transporters |
|------------------------------------------------------------------------|---------------------|---------------------|-----------------------|
| number of clusters                                                     | $59.6 \pm 2.4$      | $28.8 \pm 2.4$      | $32.4 \pm 2.4$        |
| fraction of composite units                                            | $0.76 \pm 0.03$     | $0.19 \pm 0.02$     | $0.21 \pm 0.02$       |
| ARI clusters vs hidden groups                                          | $0.15 \pm 0.02$     | $0.17 \pm 0.02$     | $0.16 \pm 0.01$       |
| cluster purity with respect to the hidden groups                       | $0.98 \pm 0.01$     | $1.00 \pm 0.00$     | $1.00 \pm 0.00$       |
| clusters per hidden group                                              | $14.9 \pm 0.6$      | $7.2 \pm 0.6$       | $8.1 \pm 0.6$         |
| raw cluster coherence $R^{\text{raw}}$                                 | $0.29 \pm 0.03$     | $0.35 \pm 0.04$     | $0.36 \pm 0.05$       |
| coverage-weighted transported coherence $R_{\text{covg}}^{\text{cov}}$ | $0.99 \pm 0.00$     | $0.80 \pm 0.02$     | $0.80 \pm 0.02$       |
| transported coherence inside the largest component                     | $0.99 \pm 0.00$     | $0.80 \pm 0.02$     | $0.80 \pm 0.02$       |
| largest-component fraction                                             | $1.00 \pm 0.00$     | $1.00 \pm 0.00$     | $1.00 \pm 0.00$       |
| strong components per candidate cluster                                | $1.00 \pm 0.00$     | $1.00 \pm 0.00$     | $1.00 \pm 0.00$       |
| mean $J/J_0$ within groups                                             | $0.93 \pm 0.01$     | $0.45 \pm 0.01$     | $0.46 \pm 0.01$       |
| mean $J/J_0$ between groups                                            | $0.04 \pm 0.01$     | $0.02 \pm 0.00$     | $0.02 \pm 0.00$       |
| median cycle-basis residual (rad)                                      | $0.0000 \pm 0.0000$ | $0.7495 \pm 0.0571$ | $0.6885 \pm 0.0518$   |
| 95th percentile cycle-basis residual (rad)                             | $0.0002 \pm 0.0000$ | $1.7064 \pm 0.1177$ | $1.7020 \pm 0.0672$   |
| maximum cycle-basis residual (rad)                                     | $0.0061 \pm 0.0046$ | $2.8868 \pm 0.2176$ | $2.7516 \pm 0.3804$   |
| fraction of cycle constraints below tolerance                          | $1.000 \pm 0.000$   | $0.007 \pm 0.005$   | $0.009 \pm 0.005$     |
| fraction of candidate units in certified clusters                      | $0.976 \pm 0.040$   | $0.000 \pm 0.000$   | $0.000 \pm 0.000$     |
| mean reciprocal two-cycle defect inside clusters (rad)                 | $0.0000 \pm 0.0000$ | $1.2854 \pm 0.0174$ | $1.3031 \pm 0.0602$   |
| final s.d. of $\omega$ within groups                                   | $0.199 \pm 0.036$   | $0.202 \pm 0.011$   | $0.194 \pm 0.009$     |
| final s.d. of $\omega$ within clusters                                 | $0.019 \pm 0.002$   | $0.008 \pm 0.000$   | $0.008 \pm 0.001$     |
| s.d. of cluster-mean $\omega$ inside a group                           | $0.296 \pm 0.059$   | $0.154 \pm 0.034$   | $0.173 \pm 0.027$     |
| transporter drift inside clusters (rad per unit time)                  | $0.0038 \pm 0.0005$ | $0.0000 \pm 0.0000$ | $0.0000 \pm 0.0000$   |

Table 5: Perturbation tests applied at  $t = T_1$  to copies of the learned networks and followed for  $T_2 = 400$  with the learning rules active (mean  $\pm$  standard deviation). Transported coherence is multiplied by reciprocal-support coverage, so a disconnected candidate cannot score as fully coherent by retaining only one component. The frozen-transporter redraw is a physical negative control; the exact compensator is a separate algebraic control.

| Quantity                                                              | Gauge network                                 | Delay-only network                        | Quenched transporters                         |
|-----------------------------------------------------------------------|-----------------------------------------------|-------------------------------------------|-----------------------------------------------|
| offset redraw: $R^{\text{raw}}$ after                                 | $0.30 \pm 0.02$                               | $0.35 \pm 0.03$                           | $0.36 \pm 0.02$                               |
| offset redraw: coverage-weighted $R^{\text{cov}}$ after               | $0.99 \pm 0.00$                               | $0.53 \pm 0.03$                           | $0.54 \pm 0.02$                               |
| offset redraw: transported coherence in largest component             | $0.99 \pm 0.00$                               | $0.87 \pm 0.02$                           | $0.85 \pm 0.03$                               |
| offset redraw: largest-component fraction                             | $1.00 \pm 0.00$                               | $0.63 \pm 0.02$                           | $0.66 \pm 0.03$                               |
| offset redraw: minimum $R^{\text{raw}}$ during                        | $0.27 \pm 0.02$                               | $0.30 \pm 0.02$                           | $0.32 \pm 0.02$                               |
| offset redraw: time to 90% recovery                                   | $89 \pm 177$                                  | $144 \pm 131$                             | $151 \pm 182$                                 |
| offset redraw: ARI of clusters after vs before                        | $0.99 \pm 0.01$                               | $0.10 \pm 0.03$                           | $0.07 \pm 0.02$                               |
| offset redraw: minimum $R^{\text{cov}}$ during                        | $0.31 \pm 0.01$                               | $0.38 \pm 0.02$                           | $0.37 \pm 0.02$                               |
| offset redraw: time to 90% covariant recovery                         | $30 \pm 2$                                    | –                                         | –                                             |
| offset redraw: cluster purity after                                   | $0.98 \pm 0.01$                               | $1.00 \pm 0.00$                           | $1.00 \pm 0.00$                               |
| offset redraw: fraction of clusters with $R^{\text{raw}} < 0.5$ after | $0.86 \pm 0.03$                               | $0.78 \pm 0.05$                           | $0.81 \pm 0.04$                               |
| offset redraw: fraction of clusters with $R^{\text{cov}} < 0.5$ after | $0.00 \pm 0.00$                               | $0.48 \pm 0.08$                           | $0.44 \pm 0.09$                               |
| exact compensator: maximum connection deviation                       | $7.2 \times 10^{-15} \pm 2.9 \times 10^{-17}$ | $5.6 \times 10^{-16} \pm 0.0 \times 10^0$ | $1.8 \times 10^{-15} \pm 1.6 \times 10^{-17}$ |
| offset redraw with learned transporters frozen: $R^{\text{cov}}$      | $0.41 \pm 0.01$                               | –                                         | –                                             |
| offset redraw with learned transporters frozen: ARI                   | $0.02 \pm 0.00$                               | –                                         | –                                             |
| removal: $R^{\text{raw}}$ of survivors                                | $0.33 \pm 0.02$                               | $0.41 \pm 0.04$                           | $0.40 \pm 0.05$                               |
| removal: $R^{\text{cov}}$ of survivors                                | $0.85 \pm 0.04$                               | $0.72 \pm 0.04$                           | $0.71 \pm 0.01$                               |
| removal: fraction of clusters with $R^{\text{raw}} < 0.5$             | $0.82 \pm 0.05$                               | $0.66 \pm 0.10$                           | $0.70 \pm 0.07$                               |
| removal: fraction of clusters with $R^{\text{cov}} < 0.5$             | $0.08 \pm 0.04$                               | $0.19 \pm 0.06$                           | $0.23 \pm 0.05$                               |
| removal: largest-component fraction                                   | $0.86 \pm 0.04$                               | $0.83 \pm 0.03$                           | $0.80 \pm 0.01$                               |
| frequency step: minimum $R^{\text{raw}}$ during                       | $0.22 \pm 0.02$                               | $0.25 \pm 0.02$                           | $0.25 \pm 0.01$                               |
| frequency step: $R^{\text{raw}}$ end / before                         | $1.10 \pm 0.32$                               | $0.97 \pm 0.22$                           | $1.07 \pm 0.06$                               |
| frequency step: re-locking time                                       | $18 \pm 9$                                    | $20 \pm 17$                               | $13 \pm 1$                                    |
| frequency step: residual $\omega$ offset                              | $0.16 \pm 0.01$                               | $0.30 \pm 0.04$                           | $0.33 \pm 0.05$                               |

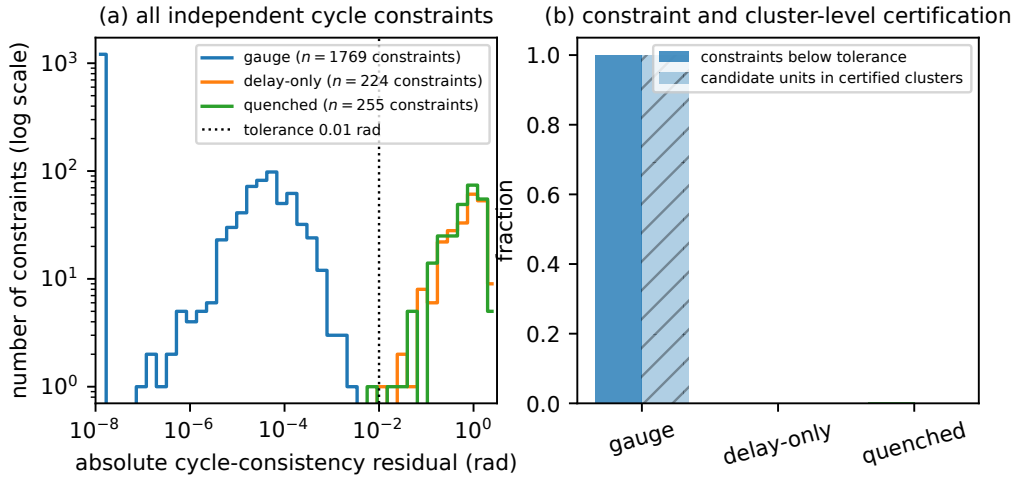


Figure 3: Complete connection-consistency diagnostics for one realisation. (a) Absolute residuals on every reciprocal two-cycle and on a fundamental basis of all longer cycles in each candidate. The dotted line is the 0.01-rad certification tolerance. (b) Fraction of individual cycle constraints below tolerance and fraction of candidate units belonging to clusters for which every tested constraint passes. This test includes triangle-free subgraphs and therefore does not infer flatness from triangles alone.

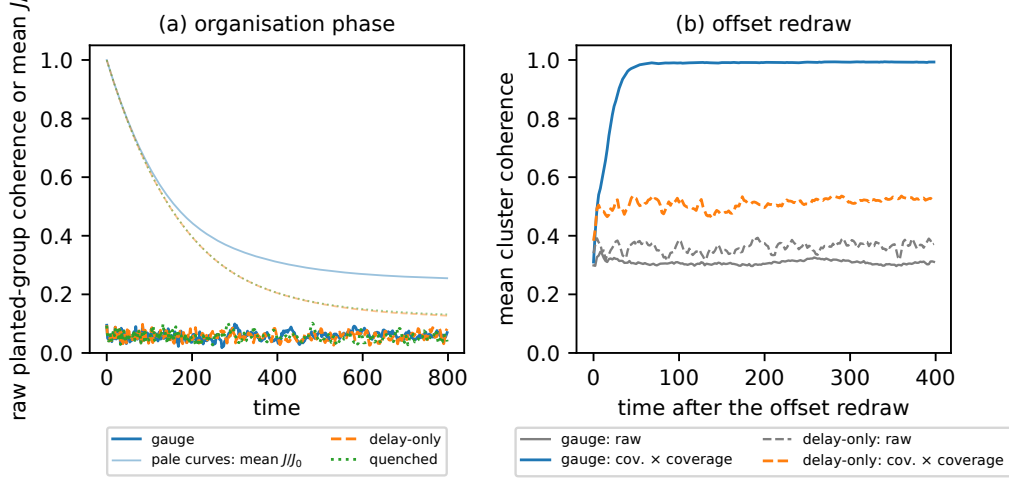


Figure 4: Time series for the realisation in Figures 2 and 3. (a) Raw coherence of the planted groups and mean coupling during organisation. (b) Raw and coverage-weighted transported coherence of the pre-redraw candidates after every physical transmission offset is redrawn.

in 32.4 candidates for quenched transporters. Candidate purity with respect to the unobserved planted groups is 0.98, 1.00 and 1.00, respectively. The adjusted Rand indices are 0.15, 0.17 and 0.16; together with the reported clusters per planted group, these numbers distinguish recruitment from recovery of the planted partition.

The coverage-weighted transported coherence is 0.99 for the gauge arm, 0.80 for delay only and 0.80 for quenched transporters. The corresponding coherence restricted to the largest reciprocal-support component is 0.99, 0.80 and 0.80, while the respective coverages are 1.00, 1.00 and 1.00. Reporting both quantities prevents a small coherent component from standing in for a disconnected candidate. Raw laboratory-frame coherence is reported separately in Table 4; it is not invariant under the phase convention of Section 4.

### *7.2. P1: complete cycle consistency*

P1 is tested on a complete independent set of cycle constraints rather than on triangles alone. For the gauge arm the median, 95th percentile and maximum absolute residual are 0.0000, 0.0002 and 0.0061 rad. The delay-only values are 0.7495, 1.7064 and 2.8868 rad, and the quenched-transporter values are 0.6885, 1.7020 and 2.7516 rad. The fractions of constraints below 0.01 rad are 1.000, 0.007 and 0.009; the fractions of candidate units lying in clusters for which every tested constraint passes are 0.976, 0.000 and 0.000. The reciprocal two-cycle defects are reported independently in Table 4. These are the quantities that decide whether Proposition 3 applies to a learned candidate.

P1 is confirmed in its stated comparative form. The adaptive-transporter arm has the smaller 95th-percentile residual and the larger certified-unit fraction than either control. Certification is not universal: 0.976 of its candidate units belong to clusters in which every tested constraint is below tolerance, and its maximum residual is 0.0061 rad. Candidate and certified cluster therefore remain distinct categories. The smaller-step run in Appendix Table B.12 is retained as a discretisation check because strict certification is necessarily conditional on the stated tolerance and numerical resolution.

### *7.3. P2: recruitment and recovery of the planted partition*

P2 has two separately testable parts. Recruitment is measured by the candidate fraction and candidate sizes. Recovery of the planted grouping is measured by the adjusted Rand index and is interpreted together with purity and the number of candidates per planted group. The gauge-to-delay-only ratio of the assigned fractions is 4.0, while the ratio of candidates per

planted group is 2.1. The numerical values above decide whether the adaptive transporter recruits more units, whether it merges or fragments planted groups, and whether any increase in recruitment actually improves partition recovery.

The recruitment part of P2 is confirmed: the gauge arm assigns 4.0 times the delay-only fraction of units to candidates. The partition-recovery part fails its pre-stated directional criterion in this campaign. Its adjusted Rand index, 0.15, is not higher than the delay-only value 0.17 or the quenched value 0.16; the paired gauge-minus-delay difference across matched realisations is  $-0.022 \pm 0.034$ . Purity remains high, but the gauge arm forms 14.9 candidates per planted group compared with 7.2 and 8.1 in the controls. Adaptive transport therefore recruits more units while fragmenting, rather than recovering, the planted partition more faithfully. This finite campaign rejects the declared improvement criterion; it does not establish that no other graph or parameter regime can improve alignment.

#### *7.4. Held-out second-scale extension: information proposes, gauge transport certifies*

The negative fine-scale P2 verdict is not changed after the fact. It establishes that local adaptive transport does not infer which pure fragments belong to one larger planted group when the sparse fine graph does not carry that comparison. The held-out experiment of Section 6.9 asks the separate constructive question: after certified fragments are promoted, what happens when an observable next-scale relation supplies the missing comparisons?

The post-run diagnostic confirms that the observable descriptor carries scale information: mean-frequency affinity is 0.907 for fragments from the

Table 6: Held-out two-level experiment on seeds 10–14 (mean  $\pm$  standard deviation over 5 realisations). “Selected” denotes the label-free cycle-certificate rule fixed in the development pilot. The fixed- $k = 4$  frequency, random-link and planted-label rows have matched added-link counts. Planted labels are used only for the reported ARI and purity and for the explicitly labelled oracle.

| <i>(a) Partition recovery</i>           |                     |                 |                   |                   |
|-----------------------------------------|---------------------|-----------------|-------------------|-------------------|
| Construction                            | assigned            | modules         | ARI               | purity            |
| Fine certified (level 0)                | $0.757 \pm 0.019$   | $57.8 \pm 4.0$  | $0.163 \pm 0.071$ | $0.979 \pm 0.003$ |
| Selected gauge (selected)               | $0.757 \pm 0.019$   | $23.6 \pm 26.8$ | $0.558 \pm 0.397$ | $0.915 \pm 0.082$ |
| Selected control (same k)               | $0.757 \pm 0.019$   | $26.4 \pm 28.9$ | $0.521 \pm 0.374$ | $0.941 \pm 0.085$ |
| Frequency gauge (fixed k=4)             | $0.757 \pm 0.019$   | $4.4 \pm 0.5$   | $0.763 \pm 0.121$ | $0.892 \pm 0.075$ |
| Frequency control (fixed k=4)           | $0.757 \pm 0.019$   | $6.2 \pm 1.5$   | $0.743 \pm 0.105$ | $0.927 \pm 0.082$ |
| Random gauge (fixed k=4)                | $0.757 \pm 0.019$   | $20.2 \pm 3.3$  | $0.444 \pm 0.098$ | $0.894 \pm 0.062$ |
| Label-oracle gauge (fixed k=4)          | $0.757 \pm 0.019$   | $4.4 \pm 0.5$   | $0.763 \pm 0.121$ | $0.892 \pm 0.075$ |
| <i>(b) Complete-cycle certification</i> |                     |                 |                   |                   |
| Construction                            | cycle $q_{0.95}$    |                 | certified         |                   |
| Fine certified (level 0)                | –                   |                 | 1.000             |                   |
| Selected gauge (selected)               | $0.0003 \pm 0.0003$ |                 | $1.000 \pm 0.000$ |                   |
| Selected control (same k)               | $0.6457 \pm 0.5885$ |                 | $0.407 \pm 0.541$ |                   |
| Frequency gauge (fixed k=4)             | $0.0128 \pm 0.0149$ |                 | $0.737 \pm 0.266$ |                   |
| Frequency control (fixed k=4)           | $0.8215 \pm 0.2185$ |                 | $0.021 \pm 0.018$ |                   |
| Random gauge (fixed k=4)                | $0.0106 \pm 0.0121$ |                 | $0.822 \pm 0.131$ |                   |
| Label-oracle gauge (fixed k=4)          | $0.0128 \pm 0.0149$ |                 | $0.737 \pm 0.266$ |                   |

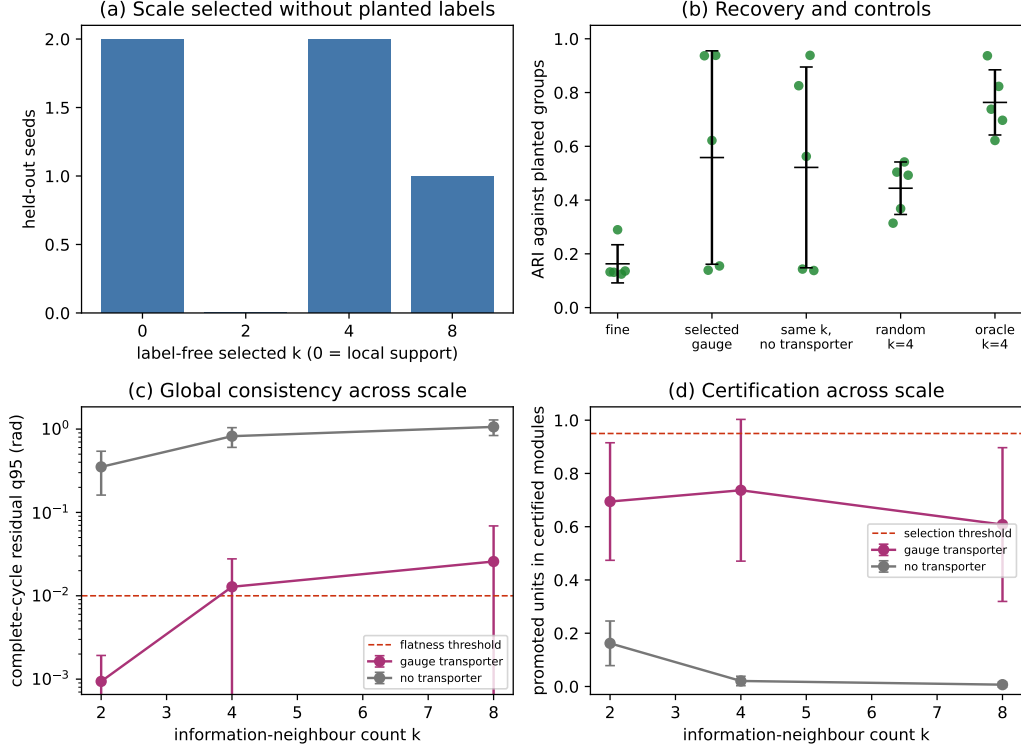


Figure 5: Held-out next-scale results. The selector may abstain and retain local boundary support ( $k = 0$ ). Partition scores show what the observable frequency support contributes; complete-cycle residuals and certified fractions show whether the proposed pairwise relations glue into a path-independent composite. The full  $k = 2, 4, 8$  sensitivity is in Appendix Table B.11.

same hidden group and 0.266 for fragments from different groups. At the fixed equal-count control  $k = 4$ , the frequency-informed gauge construction reaches ARI  $0.763 \pm 0.121$ , compared with  $0.163 \pm 0.071$  for the certified fine fragments and  $0.444 \pm 0.098$  for random added links. The paired gains are  $0.600 \pm 0.177$  and  $0.319 \pm 0.172$ . In this synthetic benchmark the frequency-informed and planted-label oracle rows coincide because the planted groups were generated as separated intrinsic-frequency bands. That equality is a positive control showing that the declared descriptor contains the intended scale information; it is not evidence that gauge transport inferred an unavailable label. Thus an informative relation at the promoted scale resolves much of the fragmentation that P2 exposed.

Most of that partition gain belongs to the supplied information, not to gauge transport. On exactly the same  $k = 4$  support the no-transporter arm reaches ARI  $0.743 \pm 0.105$ ; the paired gauge increment is only  $0.020 \pm 0.091$ . The additional gauge result is global composability: its complete-cycle 95th percentile is  $0.0128 \pm 0.0149$  rad and its certified fraction is  $0.737 \pm 0.266$ , compared with  $0.8215 \pm 0.2185$  rad and  $0.021 \pm 0.018$  without transporters. Scale information says which fragments should be compared; the transporter supplies a covariant phase relation and the complete-cycle test decides whether all pairwise relations can coexist in one larger composite.

The label-free selector accepts a non-zero new scale in 3 of 5 held-out realisations and abstains in 2. Its selected- $k$  distribution is 0:2, 4:2, 8:1. Across all five realisations, including the abstentions, selected gauge ARI is  $0.558 \pm 0.397$ , compared with  $0.163 \pm 0.071$  before coarse gluing and  $0.521 \pm 0.374$  on the matched selected support without transporters. The selected modules

have cycle 95th percentile  $0.0003 \pm 0.0003$  rad and certified fraction  $1.000 \pm 0.000$ , against  $0.6457 \pm 0.5885$  rad and  $0.407 \pm 0.541$  for the matched control. The gauge-arm cycle and certification values are acceptance conditions of the selector (or inherited from the certified local fragments when it abstains), not independent evidence of selection quality; the held-out outcomes are the acceptance frequency and the unopened-label ARI reported after selection. The experiment therefore demonstrates a valid two-level construction and a conservative certificate, while the two abstentions show that automatic scale selection is not yet reliable in every realisation. Two tested levels do not establish a universal scaling law.

#### *7.5. P3: physical offset redraw and its controls*

The offset-redraw experiment changes every fixed transmission offset and is therefore a physical perturbation, not a gauge transformation. With learning active, the gauge candidates end at coverage-weighted transported coherence 0.99 and partition identity 0.99 relative to the pre-redraw candidates. The delay-only values are 0.53 and 0.10, and the quenched values are 0.54 and 0.07. Coverage after the perturbation is reported separately in Table 5.

Two controls separate distinct claims. Algebraically replacing each transporter by the old transporter minus the offset change leaves the total connection unchanged to a maximum deviation  $7.2 \times 10^{-15}$ ; this verifies Proposition 1 but says nothing about whether the learning dynamics finds that compensator. Conversely, freezing the learned transporters during the same physical redraw gives transported coherence 0.41 and partition identity 0.02. Comparing the active and frozen tests asks whether transporter adaptation, rather than merely the learned pre-redraw state, accounts for recovery.

P3 is confirmed for the declared offset-redraw observables. Active transporter learning leaves both coverage-weighted coherence and partition identity above the two controls. Freezing the same learned transporters reduces those values to 0.41 and 0.02, whereas active adaptation gives 0.99 and 0.99. The contrast attributes the observed recovery to post-redraw adaptation rather than to the pre-redraw network alone; it does not convert the perturbation into a symmetry transformation.

#### *7.6. P4: static covariance and non-vacuity controls*

An independent random rephasing of all final unit states, accompanied by the transporter transformation of Definition 4, changes every reported invariant by less than the numerical tolerance in each gauge-arm realisation. The control applies to a time-independent change of representation, which is the symmetry proved in Proposition 2; no time-dependent covariance is claimed. The transporter update is also non-zero on the positive-control state. P4 therefore tests both invariance and non-vacuity rather than accepting a formally invariant but inactive rule.

#### *7.7. Removal and frequency perturbations*

Removing 0.2 of the units leaves coverage-weighted transported coherence 0.85, 0.72 and 0.71 in the three arms, with coverage 0.86, 0.83 and 0.80. The fractions of pre-removal candidates below the coherence threshold are 0.08, 0.19 and 0.23. This perturbation is not implied by the phase symmetry and is reported as a separate empirical test.

After shifting the natural frequency of half the units in one planted group, the minimum raw coherence, final-to-initial ratio, re-locking time and resid-

ual frequency offset are reported in Table 5. A frequency step is neither a representation change nor an offset redraw, so agreement or separation between arms here diagnoses the learned coupling state rather than gauge covariance.

#### *7.8. Milder offsets and sensitivity runs*

The complete production campaign is repeated with offsets restricted to  $(-1.571, 1.571)$ . Appendix Table B.12 also reports the corrected pilot, the coupling-weighted transporter rule, a faster frequency-adaptation rate and a halved integration step. The smaller-step run is a discretisation control; the other settings show whether the conclusions depend on the transporter objective or on the separation of learning time scales. These runs were retained whether favourable or unfavourable.

#### *7.9. Outcome of the four predictions*

P1, P3 and P4 are confirmed in the comparative forms stated before production. P2 is confirmed for recruitment and fails its pre-stated alignment criterion in this finite campaign. The campaign therefore supports strong suppression of complete-cycle inconsistency, greater candidate recruitment, adaptive recovery after an offset redraw, and static covariance. It does not support the stronger claims that every candidate is a certified composite, that the planted partition is recovered more accurately, or that phase covariance implies robustness to unrelated perturbations.

## 8. Composite regularisation of a convolutional network

The agent-based model of Sections 6 and 7 is a dynamical system. This section carries the two ideas of the paper, composite clusters with mixing and a phase-covariant routing, into a trained feed-forward network, and reports what they do there. The transfer is deliberately modest and is stated in the honest terms of the machinery it uses. A trained feed-forward network has no local phase symmetry: its units carry real activations and a real rectifier, so the statement of the previous version of this work that such a network is invariant under a rephasing of its inputs was wrong and is withdrawn. What survives the transfer is (i) the composite structure as a constraint on the weights of a layer, which is a form of soft weight sharing [33, 34] with a specific two-level (cluster and mixture) geometry, and (ii) a routing layer whose read-out is invariant under a common rephasing of its inputs. We test both on CIFAR-10 [37] against a baseline with the same convolutional backbone and training protocol, and we report the numbers as they come out, including the ones that do not favour the construction.

### 8.1. From cluster elimination to a regulariser

In a layer with weight matrix  $W \in \mathbb{R}^{n_u \times n_{in}}$  the row  $w_u$  is the incoming weight vector of unit  $u$ . A group  $C_j$  of units whose rows are pulled toward a common learned vector  $\Lambda_j$  is the weight-space analogue of a composite cluster:  $\Lambda_j$  plays the role of the composite operator, and the units of  $C_j$  are its constituents. An elementary unit  $u \in E$  whose row is pulled toward a mixture of the cluster vectors is the analogue of partial compositeness, Eq. (10): its effective parameters are those of the clusters it mixes with, in

proportion to mixing weights  $\alpha_{uj} \geq 0$ ,  $\sum_j \alpha_{uj} = 1$ . The penalty is

$$\mathcal{L}_{\text{comp}}(W, \Lambda, \alpha) = \sum_j \sum_{u \in C_j} \|w_u - \Lambda_j\|^2 + \sum_{u \in E} \left\| w_u - \sum_j \alpha_{uj} \Lambda_j \right\|^2, \quad (24)$$

and the training objective is  $\mathcal{L}_{\text{task}} + \gamma \mathcal{L}_{\text{comp}}$ , minimised jointly over the network weights, the cluster vectors  $\Lambda_j$  and, in the adaptive variant, the mixing weights, which are parametrised as  $\alpha_u = \text{softmax}(a_u)$  so that they stay on the simplex. The cluster vectors are initialised as the mean rows of their clusters at initialisation and are learned thereafter. Two variants are compared: the *semi-composite* regulariser, in which the mixing weights are fixed and uniform ( $\alpha_{uj} = 1/n_c$ ), and the *adaptive semi-composite* regulariser, in which they are learned. Mechanically, Eq. (24) is a soft weight-sharing penalty [33] with learned centres; the novelty is only the second term, which ties the elementary units to a learned mixture of the cluster centres rather than to a centre of their own, and it is this term that the comparison between the two variants isolates. The previous version of this work computed the cluster centres as fixed scalars; here they are learned vectors, as the elimination of Section 5.2 requires (the composite operator is a state of the cluster, not a number).

## 8.2. Routing layers

Two routing layers are tested between the flattened convolutional features  $x \in \mathbb{R}^{n_{\text{in}}}$  and the dense layer. The *cosine routing* layer is the layer of the previous version: a real linear map  $x \mapsto x \cos(G)/\sqrt{n_{\text{in}}/2}$  whose weights are cosines of learned angles  $G \in \mathbb{R}^{n_{\text{in}} \times n_{\text{out}}}$  and are therefore bounded by one in modulus. It is included so that the earlier claim can be tested; we note that as originally coded the angles were initialised as standard normal

variables, whose cosines have mean  $e^{-1/2} \approx 0.61$ , so that all outputs were nearly the same multiple of the summed input, about  $\sqrt{n_{\text{in}}}$  times too large, and the network did not train (chance-level accuracy; the log is kept in the supplementary results). The layer is used here with angles initialised uniformly in  $[0, 2\pi)$  and the scale shown, which gives zero-mean, unit-variance outputs at initialisation; the original initialisation is available in the code as an option. This layer has no phase symmetry of any kind; it is a bounded-weight linear layer.

The *phase routing* layer is the one that actually carries a covariance. Its inputs are paired into  $m = n_{\text{in}}/2$  complex numbers  $c_j = x_j + ix_{j+m}$ , and its outputs are the moduli of a complex linear map,

$$y_k = \left| \sum_{j=1}^m w_{kj} e^{i\vartheta_{kj}} c_j \right|, \quad w_{kj} \in \mathbb{R}, \vartheta_{kj} \in [0, 2\pi), \quad (25)$$

with learned magnitudes and angles. A common rephasing  $c_j \rightarrow e^{i\alpha} c_j$  of all inputs leaves every  $y_k$  invariant, and a rephasing  $c_j \rightarrow e^{i\alpha_j} c_j$  of a single input is absorbed by  $\vartheta_{kj} \rightarrow \vartheta_{kj} - \alpha_j$ : the angles  $\vartheta_{kj}$  are transporters in the sense of Definition 5 and the modulus is the invariant read-out, exactly as  $R^{\text{cov}}$  is in the agent-based model. This is the precise and limited sense in which “gauge structure” enters the convolutional network: one layer with an invariant read-out, not an invariance of the whole network.

### 8.3. Matched interpolation of the phase read-out

The phase-versus-cosine comparison changes both the parameterisation and the nonlinearity, so it cannot identify the symmetry as the cause of an accuracy difference. We therefore hold the complex map  $s_k = \sum_j w_{kj} e^{i\vartheta_{kj}} c_j$

fixed in form and vary only its read-out,

$$y_k^{(\lambda)} = \sqrt{\frac{2}{1+\lambda} [(\operatorname{Re} s_k)^2 + \lambda(\operatorname{Im} s_k)^2] + \varepsilon}, \quad \lambda \in \{0, 0.5, 1\}. \quad (26)$$

At  $\lambda = 1$  this is the invariant modulus of Eq. (25); at  $\lambda = 0$  it is the magnitude of one fixed real projection and changes under a generic common rephasing;  $\lambda = 0.5$  is the fixed interior point. The factor  $2/(1+\lambda)$  keeps the expected squared output fixed when the two quadratures have equal variance, as they do at the rotationally balanced initialisation. The interpolation therefore does not simultaneously alter the initial activation scale. All three values are rerun through one script. Within each seed they have identical initial tensor and data-split fingerprints, mini-batch order, optimiser, regulariser, architecture, parameter count and training length. The direct controls are the relative root-mean-square changes of the routed features and logits under the same common phase rotation, together with prediction agreement. Clean test accuracy is the performance observable. The pre-stated attribution criterion is paired across seeds: the invariant endpoint must exceed the non-invariant endpoint reproducibly; otherwise the symmetry-specific explanation is not supported. This test concerns the layer’s representation symmetry and does not alter the separate adversarial-image result.

#### 8.4. Protocol

All configurations share the backbone  $\text{Conv}(3 \rightarrow 32, 3 \times 3) - \text{ReLU} - \text{MaxPool} - \text{Conv}(32 \rightarrow 64, 3 \times 3) - \text{ReLU} - \text{MaxPool} - \text{Flatten} (n_{\text{in}} = 2304) - [\text{routing layer, 64 outputs}] - \text{Dense}(64) - \text{ReLU} - \text{Dropout}(0.5)$  [38] –  $\text{Dense}(10)$ . The regulariser acts on the rows of the  $\text{Dense}(64)$  layer, with  $n_c = 4$  clusters of 10 units and 24 elementary units. The routing layer,

when present, is inserted before Dense(64), so the three regularised configurations without routing have the same number of parameters as the baseline and the two routed configurations have more; the table reports the parameter counts. The data are the CIFAR-10 training set, split into 45 000 training and 5 000 validation images with the seed, and the 10 000 test images. Training uses Adam [39] with learning rate  $10^{-3}$ , mini-batches of 64, 8 epochs, and the final-epoch model is evaluated, exactly as implemented in the supplied code. The reported quantities are the clean test accuracy on all test images; the accuracy under the fast gradient sign attack [40] and under projected gradient descent with 10 steps, step size  $\epsilon/4$  and a random start [41], both with  $\ell_\infty$  radius  $\epsilon = 0.03$  on pixels in  $[0, 1]$ , evaluated on the first 2000 test images; and the cluster compactness of the regularised layer,  $1 - \sum_j \sum_{u \in C_j} \|w_u - \bar{w}_{C_j}\|^2 / \sum_u \|w_u - \bar{w}\|^2$ , which is 1 when every cluster has collapsed to its centre. Each configuration is run with three seeds (which fix the split, the initialisation, the mini-batch order and the dropout masks) and the table reports mean and standard deviation over seeds; the regularisation strength is  $\gamma = 10^{-2}$ , with  $\gamma \in \{10^{-3}, 10^{-1}\}$  reported for the adaptive variant at seed 0. The numbers of the previous version (clean accuracies of 63.6, 67.8 and 68.3 per cent for three configurations) compared networks of different architectures and are withdrawn; all rows below share the convolutional backbone and training protocol stated above, while the routed rows contain the explicitly reported additional layer.

### 8.5. Results

Table 7 reports the five configurations and Table 8 the dependence on the regularisation strength. The absolute accuracies are those of a two-

Table 7: CIFAR-10 results (mean  $\pm$  s.d. over three seeds; adversarial accuracies on the first 2000 test images at  $\epsilon = 0.03$ ). All rows share the backbone and protocol of Section 8.4; the routed rows contain one additional layer. “Inference parameters” counts the parameters used at prediction time, whereas “total trainable” also counts the cluster centres and mixing logits used only by the regulariser during training.

| Configuration                            | seeds | inference params | total trainable | clean (%)      | FGSM (%)      | PGD-10 (%)    | compactness     |
|------------------------------------------|-------|------------------|-----------------|----------------|---------------|---------------|-----------------|
| Baseline (no regulariser)                | 3     | 167,562          | 167,562         | $61.9 \pm 1.5$ | $8.3 \pm 1.0$ | $2.2 \pm 0.3$ | $0.07 \pm 0.02$ |
| Semi-composite (fixed uniform mixing)    | 3     | 167,562          | 176,778         | $60.9 \pm 0.3$ | $7.3 \pm 0.9$ | $1.7 \pm 0.5$ | $0.98 \pm 0.01$ |
| Adaptive semi-composite (learned mixing) | 3     | 167,562          | 176,874         | $59.2 \pm 0.8$ | $6.4 \pm 1.0$ | $1.5 \pm 0.2$ | $0.98 \pm 0.00$ |
| Adaptive + cosine routing layer          | 3     | 171,658          | 172,010         | $56.2 \pm 0.5$ | $6.2 \pm 1.0$ | $1.9 \pm 0.4$ | $0.76 \pm 0.01$ |
| Adaptive + phase routing layer           | 3     | 171,658          | 172,010         | $65.7 \pm 0.4$ | $5.8 \pm 0.7$ | $1.2 \pm 0.3$ | $0.73 \pm 0.02$ |

Table 8: Dependence on the regularisation strength for the adaptive semi-composite configuration (seed 0).

| $\gamma$ | clean (%) | FGSM (%) | PGD-10 (%) | compactness |
|----------|-----------|----------|------------|-------------|
| 0.001    | 64.7      | 9.2      | 2.2        | 0.64        |
| 0.01     | 60.0      | 7.5      | 1.7        | 0.98        |
| 0.1      | 57.5      | 5.7      | 1.1        | 1.00        |

Table 9: Moment-matched interpolation of the phase read-out (mean  $\pm$  s.d. over 3 paired seeds). Within a seed, all three rows have identical initial tensors, data split, mini-batch order, optimiser, architecture, parameter count and training length; only  $\lambda$  changes. The initial routed RMS checks the activation-scale control. Route and logit defects are relative RMS changes under the same common phase rotation, and agreement is the fraction of predictions unchanged.

| $\lambda$ | initial routed RMS  | clean accuracy (%) | compactness       | route defect        | logit defect        | prediction agreement |
|-----------|---------------------|--------------------|-------------------|---------------------|---------------------|----------------------|
| 0.0       | $0.1741 \pm 0.0306$ | $64.8 \pm 1.0$     | $0.722 \pm 0.059$ | $0.4128 \pm 0.0117$ | $0.4121 \pm 0.0420$ | $0.630 \pm 0.019$    |
| 0.5       | $0.1712 \pm 0.0313$ | $65.5 \pm 0.2$     | $0.731 \pm 0.075$ | $0.1484 \pm 0.0020$ | $0.1546 \pm 0.0136$ | $0.889 \pm 0.010$    |
| 1.0       | $0.1697 \pm 0.0317$ | $64.6 \pm 1.6$     | $0.740 \pm 0.061$ | $0.0000 \pm 0.0000$ | $0.0000 \pm 0.0000$ | $1.000 \pm 0.000$    |

convolution backbone trained for 8 epochs and are far below the state of the art on CIFAR-10; every comparison below is between rows that share that backbone, and none is a comparison with the literature.

*The regulariser does what it is designed to do, and it costs accuracy.* Cluster compactness rises from 0.07 in the baseline to 0.98 with fixed uniform mixing and 0.98 with learned mixing: the rows of each cluster have collapsed onto their learned centre, so the regularised layer is genuinely described by four composite operators and a mixture, which is the weight-space form of the elimination of Section 5.2. The price is clean accuracy,  $60.9 \pm 0.3$  and  $59.2 \pm 0.8$  per cent against  $61.9 \pm 1.5$  for the baseline. Table 8 shows the trade-off directly: as  $\gamma$  goes from 0.001 to 0.1 the compactness rises from 0.64 to 1.00 while the clean accuracy falls from 64.7 to 57.5 per cent. Collapsing a layer onto a few centres is not free; this is the expected behaviour of a soft weight-sharing penalty [33, 34], and the value of the construction here is the structure it imposes, not an accuracy gain.

*Learned mixing does not pay for itself on this task.* The adaptive variant is 1.7 points below the fixed-uniform variant at the same compactness. The two differ only in the second term of Eq. (24), the term that ties an elementary unit to a learned mixture of cluster centres and is the weight-space analogue of partial compositeness; it is also the only part of the penalty that is not standard soft weight sharing. On CIFAR-10 at  $\gamma = 0.01$  that term is a net cost. We report the comparison because it isolates exactly the ingredient that is new.

*The composite structure does not confer adversarial robustness.* Under both attacks every regularised configuration is below the baseline: FGSM  $7.3 \pm 0.9$  and  $6.4 \pm 1.0$  against  $8.3 \pm 1.0$  per cent, PGD-10 1.7 and 1.5 against 2.2. The  $\gamma$  scan repeats the pattern (FGSM 9.2, 7.5, 5.7 per cent for increasing  $\gamma$ ). The claim of the previous version of this work, that composite structure confers robustness on a trained network, is therefore not supported by these experiments and is withdrawn. Two caveats stop this from being a strong negative result. First, every row is between 1 and 9 per cent under attack: an undefended network of this size is already broken at  $\epsilon = 0.03$ , and differences of one or two points on a broken network carry little information. Second, adversarial robustness in input space is a different property from the invariance this paper is about, which is an invariance of the read-out under a rephasing of the state; the robustness that we do demonstrate is the one measured in Section 7, where the assembly structure survives a complete redraw of the connection offsets.

*The matched interpolation does not attribute the clean-accuracy difference to invariance.* The adaptive network with the phase routing layer reaches  $65.7 \pm 0.4$  per cent, 3.8 points above the unregularised baseline and 6.5 points above the same regularised network without routing. The closest capacity-matched comparison is with cosine routing: both routed rows contain 171,658 inference parameters and 172,010 total trainable parameters, and they have the same input and output dimensions. The phase design is 9.5 points higher in clean accuracy. This comparison controls parameter count but does not isolate covariance alone: the two layers also differ in parameterisation and in the modulus nonlinearity of Eq. (25).

Table 9 performs the direct intervention registered in Section 8.3. The initial routed RMS differs by at most 4.9 per cent across the three values of  $\lambda$  within a seed. The transformation control separates exactly as intended: at  $\lambda = 0$  the route and logit defects are 0.4128 and 0.4121 with prediction agreement 0.630, whereas at  $\lambda = 1$  both defects round to 0.0000 and agreement is 1.000. Exact invariance is therefore implemented, but it does not produce the pre-stated performance ordering. Clean accuracy is  $64.8 \pm 1.0$ ,  $65.5 \pm 0.2$ , and  $64.6 \pm 1.6$  per cent at  $\lambda = 0$ , 0.5, and 1, respectively. The paired invariant-minus-broken difference is  $-0.2 \pm 1.3$  points; the invariant endpoint wins in only 1 of 3 seeds and the three values are monotone in only 1 seed. The symmetry-specific attribution criterion is therefore not met. The interior point has the highest mean by 0.7 points over  $\lambda = 0$  and 0.9 points over  $\lambda = 1$ , but this non-monotone interior maximum was not the directional criterion and is reported as exploratory. The complete phase-routing design remains associated with higher clean accuracy than cosine routing in the original capacity-matched comparison; this experiment shows that exact read-out invariance is not the cause identified by the present data. The cosine accuracies of the previous version remain withdrawn.

*What the routed rows do not show.* Phase routing is the least robust configuration in the table under both attacks (FGSM  $5.8 \pm 0.7$ , PGD-10  $1.2 \pm 0.3$  per cent), and no penalised or routed row improves on the unregularised baseline under either attack, so the invariant read-out does not transfer its invariance to the input space, as it should not: the invariance of Eq. (25) is an invariance under rephasing the layer’s own inputs, and an adversarial perturbation of an image is not a rephasing. The compactness column is also not

comparable across the routed and unrouted blocks, because with a routing layer the regularised layer is  $\text{Dense}(64 \rightarrow 64)$  and its rows are 64-dimensional, while without one they are  $n_{\text{in}} = 2304$ -dimensional; compactness should be read within a block.

Taken together: on this task the composite regulariser buys an interpretable two-level structure at a cost of one to three points of accuracy, the phase-routing design is associated with higher clean accuracy in the capacity-matched routing comparison, the paired interpolation does not attribute that difference to exact phase invariance, and neither construction buys adversarial robustness. We state this rather than selecting the favourable rows, because the point of the transfer is to find out which parts of the construction of Sections 4–5 survive in a setting that has no phase symmetry to begin with. The composite structure survives as a constraint, while a covariant read-out can be built by reintroducing a complex state; the matched interpolation establishes the representation property and rejects its proposed clean-accuracy attribution under this protocol.

## 9. Discussion

### 9.1. *What is established analytically*

The transporter of a directed link is, in neural language, a learnable phase lag. A static redefinition of each unit’s phase changes neither the dynamics nor any of the three local updates when the transporter is transformed with it (Proposition 2). This is a representation symmetry. It does not make a physical change of the fixed transmission offsets into a gauge transformation. Proposition 1 is instead an algebraic statement: for any offset change there

exists a compensating transporter change that leaves the total connection unchanged. Whether local learning finds that compensator is an empirical question tested by the active and frozen redraw experiments of Section 7.

Fixed random phase offsets and adaptive couplings have the structure of a gauge-glass model [25, 26], while adding dynamical link phases resembles a graph gauge-matter system. These are structural correspondences, not identifications of the neural model with an equilibrium gauge glass or a Higgs phase. The complete cycle-basis residuals give the relevant graph statement directly. A candidate is a certified composite only when every independent cycle constraint passes the tolerance; only then is the transported mean of Definition 9 path independent.

The cluster elimination has a similarly precise boundary. Its Schur complement gives exact fixed points and zero-frequency gains. At finite frequency the eliminated cluster contributes the memory kernel of Eq. (14), and its exact transient time scales are the poles of Eq. (17). Calling the static-local estimate an exact effective time constant would erase that distinction; the numerical comparison in Section 5.2 shows that the error becomes large near the stability boundary.

### *9.2. What the numerical tests establish*

The corrected agent-based experiment separates four questions: complete cycle consistency, recruitment and partition recovery, adaptation after a physical offset redraw, and static covariance. Coverage-weighted coherence prevents disconnected candidates from receiving full credit through one retained component. Reciprocal two-cycles and a fundamental cycle basis test all independent loops, including triangle-free candidates. The exact

compensator and random-rephasing checks validate the algebra, whereas the frozen-transporter redraw tests whether recovery actually requires adaptation. Section 7 reports these quantities separately and preserves failed prediction components.

The production results give the adaptive link variable a specific empirical role. It reduces the 95th-percentile complete-cycle residual from 1.7064 rad in the delay-only arm to 0.0002 rad and raises the fraction of candidate units in fully certified clusters from 0.000 to 0.976. It also recruits 4.0 times the delay-only candidate fraction, but the corresponding adjusted Rand index is 0.15 rather than 0.17: recruitment improves while recovery of the hidden partition does not. Following the offset redraw, active adaptation retains coverage-weighted coherence 0.99 and partition identity 0.99, whereas freezing the transporters gives 0.41 and 0.02. These comparisons support the link-adaptation mechanism under the declared perturbation while preserving the failed partition-alignment prediction.

The held-out hierarchy experiment identifies what was missing from that fine-scale recovery. Certified fragments cannot decide that they share a larger group unless some observable compares fragments that the sparse fine graph keeps apart. Composite mean frequency supplies such a relation here: at fixed  $k = 4$  it raises ARI from 0.163 to 0.763, while random equal-count links reach 0.444. Because the synthetic planted groups are frequency bands, this frequency support coincides with the planted-label oracle in the fixed- $k$  control; it is an explicit positive control of the supplied information rather than an inference of labels by the gauge variables. On the same frequency support, removing transporters changes ARI only from 0.763 to 0.743, but

worsens the cycle 95th percentile from 0.0128 to 0.8215 rad. Gauge transport therefore does not manufacture the scale descriptor; it supplies the covariant phases and tests whether the pairwise relations glue consistently. The label-free selector abstains in 2 of 5 runs, so robust automatic scale discovery and a universal scaling law remain open.

The word “robustness” is therefore used only with an explicit perturbation and observable. Offset-redraw recovery means retention of coverage-weighted transported coherence and partition identity after a physical change of every link offset. Removal of units and a frequency step are different perturbations and are reported independently. Input-space adversarial robustness is different again. The CIFAR-10 experiment does not support that claim: every regularised or routed row is below the unregularised baseline under both attacks, so the adversarial-robustness claim of the previous version is withdrawn.

### 9.3. Two concrete applications

*Brain-network models with transmission delays.* Whole-brain oscillator models couple regions with delays related to fibre length [42, 6]. At a specified oscillation frequency a delay contributes a phase offset. A per-link adaptive lag could represent an effective adjustment of propagation or local phase response; activity-dependent myelination is one possible biological mechanism [43]. The model yields a discriminating measurement: inside a certified assembly, reciprocal and fundamental-cycle phase residuals should all lie below the stated tolerance. Pairwise synchrony alone cannot establish that property, because a spanning-tree alignment leaves non-tree loop inconsistency untested.

*Soft weight sharing in trained networks.* The weight-space counterpart is operationally simpler. Cluster centres represent shared incoming-weight patterns, and the elementary rows are tied to mixtures of those centres. On the tested CIFAR-10 backbone the penalty produces compact centres but reduces clean and adversarial accuracy; learned mixing performs worse than fixed uniform mixing. The phase-routing design has higher clean accuracy than the capacity-matched cosine-routing design, but the two layers differ in parameterisation and nonlinearity as well as covariance. The paired, moment-matched read-out interpolation supplies the missing intervention: it verifies exact invariance at  $\lambda = 1$ , yet the invariant-minus-broken accuracy difference is  $-0.2 \pm 1.3$  points and is positive in only 1 of 3 seeds. The symmetry-specific explanation is therefore not supported. The interior point has the highest mean, an exploratory indication that angular anisotropy and optimisation may interact non-monotonically rather than through a simple more-invariance-is-better law.

#### 9.4. Limitations

The complex phase unit is a normal form near oscillation onset, not a biophysical neuron model. The link sensitivity  $q_i$  controls adaptation and is not a representation charge. The covariance proved here is static; a time-dependent local rephasing would require an additional temporal connection. Candidate membership depends on a coupling threshold, and certification depends on a declared numerical tolerance. The random geometric graph has variable realised degree and boundary effects. The Euler–Maruyama simulation is synchronous, so no asynchronous-update conclusion is drawn. The production campaign uses finite network size and a finite number of seeds,

and the smaller-step run is the available discretisation check. The hierarchy uses one synthetic scale descriptor, two levels and five held-out realisations; it does not establish universal scale invariance, and its conservative selector abstains twice. Finally, the convolutional counterpart is one small architecture on one data set for 8 epochs; its read-out interpolation uses only 3 paired seeds and isolates the declared transformation parameter only within this architecture. It tests a construction, not a general performance claim.

#### *9.5. What would be required for broader claims*

The present controls also define concrete tests for claims that this study does not make. A general scale law would require at least three promoted levels, descriptors fixed before the held-out runs, and acceptance and abstention rates measured as network size and sparsity vary. A biological claim would require phase and delay data from a specified neural system, assemblies defined independently of the fitted transporters, and out-of-sample comparison of complete-cycle residuals and perturbation recovery against delay-only and non-adaptive controls. Input-space robustness would require attack-strength curves, stronger iterative and adaptive attacks, adversarially trained matched baselines, several architectures and data sets, and evaluation conditional on examples classified correctly before attack. Universal superiority would require a prospectively specified multi-task benchmark and a compute-matched comparison. These are discriminating protocols, not evidence supplied by the present simulations.

## 10. Conclusion

The revised work keeps its central proposal—composite neural assemblies coupled through learnable phase transporters—but separates the claims that the previous version had conflated.

1. A static, unit-wise phase change is a representation symmetry of the dynamics and all three local updates (Proposition 2). The transporter rule is the gradient of a normalised covariant mismatch; the coupling and frequency rules are invariant mismatch-driven updates, not additional gradients of that same objective. Any physical offset change has an exact algebraic compensator (Proposition 1), while dynamical recovery is tested separately.
2. A strongly coupled subgraph is only a candidate composite. Certification requires every reciprocal two-cycle and every fundamental cycle-basis residual to pass a declared tolerance. Under this condition, and only under it, the transported composite operator is path independent (Proposition 3).
3. Eliminating a linear cluster gives an exact Schur complement for fixed points and static gains. Transients retain a memory kernel and two exact poles; the static-local expression is not an exact time constant. The analytical scripts verify both the static identity and the difference between the static estimate and the slow pole.
4. The agent-based campaign tests four pre-stated predictions with three arms, complete loop diagnostics, coherence weighted by coverage, static covariance and exact compensator controls, a frozen transporter control, and a smaller time-step run. The outcome of each prediction is

reported in Section 7 without converting a failed component into a weaker post hoc claim.

5. A separate held-out hierarchy test supplies observable scale information only after the fine-scale prediction fails. It recovers much of the larger partition at fixed support, while matched controls show that the information produces most of the partition gain and gauge transport produces the complete-cycle consistency. The selector’s two abstentions are retained.
6. In the feed-forward counterpart, the composite penalty produces the intended compact weight clusters but costs clean accuracy and does not improve adversarial accuracy. The phase-routing design has higher clean accuracy than the capacity-matched cosine-routing design, but the paired, moment-matched read-out interpolation gives an invariant-minus-broken difference of  $-0.2 \pm 1.3$  points and satisfies the directional criterion in only 1 of 3 seeds. Exact read-out invariance is verified; its proposed clean-accuracy attribution is not supported. The broad robustness claim of the previous manuscript is withdrawn.

In the corrected production campaign, the gauge arm reduces the 95th-percentile complete-cycle residual from 1.7064 to 0.0002 rad, assigns 0.76 of the units to candidate clusters compared with 0.19 for delay only, and retains offset-redraw coherence 0.99 and partition identity 0.99 compared with 0.53 and 0.10. Freezing the learned transporters reduces the gauge values to 0.41 and 0.02, supporting an adaptive rather than purely algebraic recovery mechanism. P1, P3 and P4 are confirmed; P2 is confirmed for recruitment and fails its pre-stated fine-scale alignment criterion. In the distinct held-

out extension, observable frequency support raises fixed- $k$  ARI from 0.163 to 0.763, while the matched no-transporter score 0.743 and the cycle residuals show that scale information recovers the grouping and gauge transport chiefly certifies its global consistency. Not every candidate is certified, the scale selector abstains in 2 of 5 runs, no adversarial-robustness benefit is observed in the CNN experiment, and increasing read-out invariance does not reproduce the clean-accuracy ordering required by the pre-stated attribution test.

The resulting claim is narrower and testable. Local phase transport provides a representation-covariant way to compare and adapt oscillator states; complete loop consistency determines whether a candidate supports a well-defined composite variable; and an observable scale relation can propose comparisons between already certified composites. The gauge construction checks and adapts the phases of those relations; it does not create the missing information. Physical robustness and scale selection remain empirical properties of specified tests, not consequences inferred from the word “gauge.”

## 11. Code and data availability

The complete code, the raw result files from which every number and figure of this article is generated, and a README with the run order are provided as supplementary material (`si_code.zip`). The analytical checks for the Schur-complement reduction, covariance, complete-cycle certification and non-vacuity of the learning rule are included as scripts that print PASS/FAIL. The agent-based simulations require no input outside the unpacked folder. CIFAR-10 is obtained from the recorded public source and

checked against the supplied file manifest. A public, tagged copy of the reconstruction record is available at <https://github.com/apatrascu-cmd/composite-neuron-gauge-connectivity>.

## Appendix A. Proofs and analytical checks

*Proposition 2.* For time-independent  $\alpha_i$ ,  $z_i \rightarrow e^{i\alpha_i} z_i$  and  $\theta_{ij} \rightarrow \theta_{ij} + \alpha_i - \alpha_j$  imply  $U_{ij} \rightarrow e^{i\alpha_i} U_{ij} e^{-i\alpha_j}$ . Hence  $U_{ij} z_j$  and every covariant difference transform at  $i$ , while  $m_{ij} = \bar{\hat{z}}_i U_{ij} \hat{z}_j$  is invariant. Equation (1) is therefore covariant and the right-hand sides of Eqs. (4)–(6) are invariant functions of  $m$ ,  $J$  and  $q$ . The transformed transporter satisfies the same differential equation because  $\alpha_i - \alpha_j$  is constant. The phase increments telescope on every closed walk, so holonomies and all functions of the cycle residuals are invariant. Tree-transported states acquire one common root factor, which disappears after taking a modulus; multiplying by the membership coverage preserves invariance. The deterministic supplementary test also shows that arbitrary time-dependent rephasing is *not* a symmetry of the full adaptive system.  $\square$

*Proposition 1.* After  $\delta_{ij} \rightarrow \delta_{ij} + \Delta_{ij}$  and  $\theta_{ij} \rightarrow \theta_{ij} - \Delta_{ij}$ , the sum  $\theta_{ij} + \delta_{ij}$ , and therefore  $U_{ij}$  on every link, is exactly unchanged. The complete vector field and every diagnostic are thus identical. This paired algebraic operation is distinct from the physical offset-redraw test, which changes  $\delta$  first and asks whether the subsequent dynamics recovers.  $\square$

**Lemma 4** (Equal loop holonomies imply a static gauge difference). *On a connected reciprocal graph, suppose two directed connections  $\vartheta$  and  $\vartheta'$  have*

equal holonomy on every directed closed walk, including reciprocal two-cycles. Then there are node phases  $\beta_i$  such that  $\vartheta'_{ij} - \vartheta_{ij} = \beta_i - \beta_j$  on every directed link  $i \leftarrow j$ .

*Proof.* Choose a root and an orientation of a spanning tree away from it. Set  $\beta_{\text{root}} = 0$ . For a tree edge whose child is  $i$  and parent is  $j$ , define  $\beta_i = \beta_j + (\vartheta'_{ij} - \vartheta_{ij})$  using the directed link  $i \leftarrow j$ . Equal holonomy of the reciprocal two-cycle fixes the reverse link. Any non-tree directed link closes a fundamental directed walk with the tree and reciprocal links; equality of that walk's holonomy fixes its difference to  $\beta_i - \beta_j$ . The fundamental cycles and two-cycles span all directed closed-walk constraints. The sign agrees with the transformation law in Definition 5.  $\square$

*Proposition 3.* Two directed paths from  $k$  to  $i$ , together with the reciprocal of one path, form a closed directed walk. Their transport factors differ by the exponential of that walk's holonomy. If every closed-walk holonomy vanishes, all path transports agree. Conversely, a non-zero holonomy supplies two paths with different transport, so the transported term for a generic non-zero  $z_k$  and hence the operator depend on the path. The product along a path transforms only at its endpoints, so  $U_{ik}^{(\gamma)} z_k \rightarrow e^{i\alpha_i} U_{ik}^{(\gamma)} z_k$ .  $\square$

*Static and transient elimination.* At a fixed point, partition Eq. (8) into retained variables  $E$  and cluster variables  $C$ . With  $J_{CC} = \Lambda(\mathbf{1}\mathbf{1}^\top - I)$ ,  $J_{EC} = \epsilon\mathbf{1}^\top$  and  $J_{CE} = \mathbf{1}\epsilon^\top$ ,

$$J_{EC}(I - J_{CC})^{-1}J_{CE} = \frac{n}{1 - (n-1)\Lambda} \epsilon\epsilon^\top,$$

which proves Eq. (10). For transients, the integrating-factor solution of Eq. (9) is Eq. (14); its Laplace transform yields Eq. (15). Diagonalising

Eq. (16) gives Eq. (17); its determinant vanishes at Eq. (19). The supplementary script verifies the fixed-point identities on random matrices, compares the exact pole with the static-local and first-order low-frequency approximations, and writes every value quoted in Section 5.2.  $\square$

## Appendix B. Simulation parameters and sensitivity checks

Table B.10: Corrected production parameters, read directly from the result JSON. The support is a reciprocal random geometric graph;  $k$  is the target interior degree, not a fixed in-degree.

| Parameter                                      | Meaning                                                       | Value              |
|------------------------------------------------|---------------------------------------------------------------|--------------------|
| $N, k$                                         | units, target geometric-graph degree                          | 1000, 12           |
| $G, \Delta\Omega, \sigma_\omega$               | planted groups, spacing, initial spread                       | 4, 2, 0.05         |
| $\mu, J_0, D$                                  | Stuart–Landau growth, maximal coupling, noise                 | 1, 0.3, 0.005      |
| $\eta_\theta, \eta_J, \eta_\omega$             | transporter, coupling and frequency rates                     | 0.05, 0.005, 0.005 |
| $\delta_{\max}$                                | directed offsets uniform on $[-\delta_{\max}, \delta_{\max}]$ | $\pi$              |
| $\Delta t, T_1, T_2$                           | step, organisation interval, test interval                    | 0.05, 800, 400     |
| $n_{\min}$ , coupling threshold                | minimum candidate size, fraction of $J_0$                     | 5, 0.5             |
| $\varepsilon_\Phi$                             | maximum residual for cluster-level certification              | 0.01               |
| removal fraction, $\Delta\omega_{\text{step}}$ | perturbation magnitudes                                       | 0.2, 0.5           |
| seeds                                          | independent graphs, offsets, initial states and noise         | 5                  |

Table B.11: Held-out next-scale sensitivity to the symmetric frequency-neighbour count  $k$ . Gauge and no-transporter arms use exactly the same support for each row. The label-free selector uses these cycle diagnostics without seeing the planted labels; the ARI and purity columns are opened only for evaluation.

| $k$ | arm               | ARI               | purity            | cycle $q_{0.95}$    | certified         |
|-----|-------------------|-------------------|-------------------|---------------------|-------------------|
| 2   | gauge transporter | $0.622 \pm 0.122$ | $0.905 \pm 0.079$ | $0.0009 \pm 0.0010$ | $0.695 \pm 0.221$ |
| 2   | no transporter    | $0.541 \pm 0.088$ | $0.927 \pm 0.082$ | $0.3519 \pm 0.1904$ | $0.162 \pm 0.084$ |
| 4   | gauge transporter | $0.763 \pm 0.121$ | $0.892 \pm 0.075$ | $0.0128 \pm 0.0149$ | $0.737 \pm 0.266$ |
| 4   | no transporter    | $0.743 \pm 0.105$ | $0.927 \pm 0.082$ | $0.8215 \pm 0.2185$ | $0.021 \pm 0.018$ |
| 8   | gauge transporter | $0.630 \pm 0.294$ | $0.762 \pm 0.201$ | $0.0258 \pm 0.0431$ | $0.608 \pm 0.289$ |
| 8   | no transporter    | $0.883 \pm 0.111$ | $0.935 \pm 0.096$ | $1.0619 \pm 0.2253$ | $0.007 \pm 0.009$ |

Table B.12: Corrected-model pilot and sensitivity checks. “Coverage” is the largest connected fraction of a tracked candidate; “cycle q95”, “constraints pass” and “certified units” use the complete two-cycle plus fundamental-cycle test. The energy-rule, faster-frequency and halved-step rows change one modelling or numerical choice from the pilot.

| run              | rule       | $\Delta t$ | $\eta_\theta$ | $\eta_\omega$ | arm                   | composite | $R_{\text{cov}}^{\text{cov}}$ | coverage | cycle q95 | constraints pass | certified units | ARI redraw | frozen $R_{\text{cov}}^{\text{cov}}$ |
|------------------|------------|------------|---------------|---------------|-----------------------|-----------|-------------------------------|----------|-----------|------------------|-----------------|------------|--------------------------------------|
| pilot_v2         | normalised | 0.05       | 0.05          | 0.005         | Gauge network         | 0.69      | 0.96                          | 1.00     | 0.0045    | 0.983            | 0.659           | 1.00       | 0.40                                 |
| pilot_v2         | normalised | 0.05       | 0.05          | 0.005         | Delay-only network    | 0.24      | 0.84                          | 1.00     | 1.6416    | 0.034            | 0.000           | 0.25       | –                                    |
| pilot_v2         | normalised | 0.05       | 0.05          | 0.005         | Quenched transporters | 0.17      | 0.84                          | 1.00     | 1.6081    | 0.000            | 0.000           | 0.12       | –                                    |
| scan_energy_v2   | energy     | 0.05       | 0.05          | 0.005         | Gauge network         | 0.56      | 0.96                          | 1.00     | 0.4506    | 0.322            | 0.000           | 0.84       | 0.51                                 |
| scan_energy_v2   | energy     | 0.05       | 0.05          | 0.005         | Delay-only network    | 0.24      | 0.84                          | 1.00     | 1.6416    | 0.034            | 0.000           | 0.25       | –                                    |
| scan_etaomega_v2 | normalised | 0.05       | 0.05          | 0.01          | Gauge network         | 0.65      | 0.93                          | 1.00     | 0.0103    | 0.945            | 0.552           | 1.00       | 0.45                                 |
| scan_etaomega_v2 | normalised | 0.05       | 0.05          | 0.01          | Delay-only network    | 0.16      | 0.92                          | 1.00     | 1.5684    | 0.051            | 0.000           | 0.32       | –                                    |
| scan_dt025_v2    | normalised | 0.025      | 0.05          | 0.005         | Gauge network         | 0.66      | 0.96                          | 1.00     | 0.0024    | 0.991            | 0.854           | 0.86       | 0.41                                 |
| scan_dt025_v2    | normalised | 0.025      | 0.05          | 0.005         | Delay-only network    | 0.07      | 0.88                          | 1.00     | 1.6729    | 0.105            | 0.000           | 0.66       | –                                    |

## Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used OpenAI Codex to improve the language and readability of the manuscript. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the article.

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